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Energy poverty and gender equality in education: Unpacking the transmission channels

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ABSTRACT

Access to energy is widely known to promote socio-economic development; however, the linkage between access to energy and gender equality in education and the channels through which energy poverty affects gender equality in education is not explored much in the empirical literature. We, therefore, examine whether access to electricity and clean cooking fuels and technologies contribute to gender equality in education for a panel of 98 countries between 2000 and 2021. Using the two-step generalized method of moments technique, we found that access to clean cooking fuels and technologies is associated with improvement in the gender parity index for secondary and tertiary education enrolment. Electrification was found to relate to improvement in the gender parity index for primary, secondary, and tertiary education. Further evidence showed that rural and urban access to clean cooking fuels and technologies and rural and urban electrification enhance gender equality in education. We identified that female literacy, female health, female employment, information, and communication technologies are the potential transmission channels through which access to electricity and clean cooking fuels and technologies could contribute to gender equality at all levels of education. Our findings are robust across alternative econometric estimators.

1. Introduction

Achieving gender equality in education remains critical in sustaining economic development. Previous literature has documented that gender equality in education is associated with higher economic growth (for instance, Bandiera and Natraj, 2013; Karoui and Feki, 2018; Klasen and Lamanna, 2009). Despite efforts and the crusade to close gender inequality in education, it remains a persistent problem in developing countries. We argue that gender inequality in education is an ethical and moral concern that requires policymakers, especially in developing countries, to institute a pragmatic approach to tackle it. Most of the previous literature has attributed gender inequality in education to political/institutional, cultural, religious, and macroeconomic factors (Becker and Woessmann, 2008; Brown, 2004; Cooray and Potrafke, 2011). However, these empirical studies paid less attention to the role of access to energy in gender equality in education. The policy-relevant question that guides this study is, "*Can access to energy serve as a policy instrument for achieving gender equality in education?*" The Sustainable Energy for All (SE4All) initiative championed by the United Nations calls on all (business organizations, governments, and civil societies) to work together to ensure that access to modern and sustainable energy for all is a reality. Despite the great strides made over the years towards SE4All, many people in the world are still energy-poor (FAO, 2022; IEA, 2020b; IEA, IRENA, UNSD, WorldBank, and WHO, 2022); that is, they lack access to modern, affordable and sustainable energy (Ngarava et al., 2022). Though energy poverty is challenging for all, in developing countries, women are considered relatively more disadvantaged as their access to energy is further limited by gender disparities (Robinson, 2019; Winther et al., 2020).

In view of that women are considered mostly affected by energy

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poverty and their unparalleled role played in securing energy for their households in developing countries, the discussion on the importance of energy discreetly assumes that access to modern energy benefits women the most (Fingleton-Smith, 2018; Rewald, 2017; Ssennono et al., 2021). This is the case as access to modern energy saves women from the health and safety risks associated with using traditional biomass to meet household energy needs (Abbas et al., 2020; Kaygusuz, 2011; Robinson, 2019; Sovacool, 2012). The use of modern energy also saves women and girls a great deal of time that they could invest in other activities, including education and paid employment (Ngarava et al., 2022; Nguyen and Su, 2021; Ozughalu and Ogwumike, 2018; Pachauri and Rao, 2013; Ssennono et al., 2021). The potential effect of energy poverty generally hinges on the time burden on women and girls in providing energy for their households. This is usually mentioned as the reason why women's and girls' education suffers relatively more if households rely on biomass energy (Nguyen and Su, 2021; Porcaro and Tadaka, 2005; Rewald, 2017).

The United Nations Educational, Scientific and Cultural Organization (UNESCO) estimates that about 118.5 million girls are out of school (UNESCO, 2022). One of the major reasons for this outcome is the disproportionate allocation of household chores to girls and women, especially in developing and poor countries (Abbas et al., 2020; WorldBank, UNESCO, and UNICEF, 2021). Chunk of these household chores are energy-related; either the search/collection of energy or duties that access to modern energy would help execute faster and easier to save time that could be invested in schooling. Hence, the lack of modern energy, i.e., energy poverty, can have a detrimental effect on girls' education (Ozughalu and Ogwumike, 2018). The benefits of education to women and girls and society are enormous; education increases women's agency and decision-making, makes them contribute to economic growth, reduces teenage pregnancy, reduces maternal mortality, reduces excessive population growth, helps them find decent jobs (Dutta et al., 2017; WorldBank et al., 2021).

The gendered benefits of energy have attracted myriad empirical studies examining the lack of or inadequacy of modern energy access and some gendered factors. Despite the plethora of studies conducted, a chunk of the energy-gender literature has focused on women's empowerment and participation in household decision-making (Ding et al., 2019; Fingleton-Smith, 2018; Musango et al., 2020; Sahrakorpi and Bandi, 2021; Winther et al., 2020; Winther et al., 2018), the incidence of energy poverty and who is more energy poor (men versus women) (Heredia et al., 2022; Ozughalu and Ogwumike, 2018; Sánchez-Guevara Sánchez et al., 2020; Ssennono et al., 2021), measurements and determinants of energy poverty (Nussbaumer et al., 2012; Pachauri and Spreng, 2011), women public health factors (Pan et al., 2021), and women saving of money and time with access to modern energy (Ding et al., 2014).

A review of the extant literature reveals that not much attention has been given to the empirical analysis of the effect of energy poverty on gender equality in education. Related studies have examined how access to electricity or modern energy improves academic performance and school attendance of girls/women (Barkat et al., 2002; Brew-Hammond and Crole-Rees, 2004; Clancy, 2002; Karlsson and McDade, 2001; Lambrou and Piana, 2006; Ngarava et al., 2022). Such studies do not clearly elucidate gender inequality in education; at best only generally tell us that reduction in energy poverty improves girls'/women's performance and participation in school. However, gender inequality in education explains expansion in girls'/women's education achievement relative to boys/men and enables one to determine parity. With this we are able to ascertain whether the gender inequality in education gap is widening or improving.

To understand the importance of energy to gender inequality issues, Nguyen and Su (2021) used a sample of 51 developing countries to examine the impact of energy poverty on gender inequality issues such as gender inequality in employment and payment, gender inequality in health, gender inequality in socio-political rights and gender inequality in education. While the work of Nguyen and Su (2021) is very important, it does not offer an in-depth analysis of the channels through which energy poverty affects gender inequality. We argue that the impact of energy poverty on gender inequality, especially in education, may not always be direct but indirect as well. Therefore, the failure to empirically identify the mechanisms or channels through which energy poverty affects gender inequality in education could obscure policymakers' understanding of how energy poverty affects gender inequality in education and render energy policies to support gender equality ineffective. This knowledge gap indicates that further empirical research is needed to supplement the work of Nguyen and Su (2021) by identifying the potential causal channels through which energy poverty affects gender inequality in education, and this motivates our study. As presented in the literature section, we conceptualize and argue that information and communication technology, employment and financial security, health, literacy and academic performance are some of the potential channels through which energy poverty affects gender equality in education. This study, therefore, test empirically if these variables indeed mediate the impact of energy poverty on gender equality in education. Understanding the potential channels through which energy poverty affects gender inequality in education is important for supporting policies that seek to enhance gender equality education. Also, while women/girls are vulnerable to energy poverty, understanding and addressing the consequences of energy poverty is important for designing an inclusive and equitable educational system that benefits women or girls.

Given the identified research gap, this study examines how energy poverty contributes to gender inequality in education. Specifically, we look at how access to modern energy (electricity and clean cooking fuels and technologies) impacts girls'/women's education relative to boys'/men's. The uniqueness and contributions of this paper to the literature on energy poverty and gender inequality are discussed as follows. First, unlike the study of Nguyen and Su (2021), which used 51 developing countries, our study differs in this aspect by purposely using 98 countries from developing regions such as sub-Saharan Africa, South Asia and Latin America and the Caribbean, which are known to have higher disparity in gender equality in education (Cooray and Potrafke, 2011) and also considered energy poor (Acheampong et al., 2021b) to examine the impact of energy poverty on gender inequality in education. Using a large sample of countries is important for improving the reliability of the results while ensuring easy generalization of the findings. Also, using a larger sample of countries is more useful for formulating global policy.

Second, unlike the study of Nguyen and Su (2021), which considered only the direct impact of energy poverty on gender inequality, including gender inequality in education, our study extends and contributes to the literature by examining the possible causal channels through which access to energy could affect female education. We consider examining the channels more important than drawing a direct link between energy and education. We identified female literacy, female health, female employment, information, and communication technologies as the potential transmission channels. To the best of our knowledge, our study is the first to perform this analysis. Third, as most studies have considered mainly the impact of access to electricity, we look at how access to electricity (accounting for national, urban, and rural access) and clean cooking fuels and technologies (accounting for national, urban, and rural access) impact gender inequality in education.

Finally, this study is policy-relevant as we focus on how energy poverty affects gender inequality in education. The use of gender inequality enables us to gauge the extent of gender parity in education rather than just focusing on school attendance and the number of girls/women going to school. We consider this important as the Sustainable Development Goals (SDGs) agenda is on reducing inequality and granting all equal opportunities rather than merely increasing the number of female students. For example, Goal 4 aims at providing quality education for all and, among other targets, seeks to ensure "equal access for all women and men to affordable … education …" and that

"… all girls and boys complete … quality primary and secondary education …". Goal 5 aims to "achieve gender equality and empower all women and girls" with major targets such as ensuring "… promotion of gender equality and the empowerment of all women and girls at all levels", "ending all forms of discrimination against all women and girls everywhere" and "undertaking reforms to give women equal rights … in accordance with national laws". Goal 10 aims to reduce inequality within and among countries and specifically seeks to "… empower and promote the social, economic and political inclusion of all, irrespective of… sex …".[1] All these goals seek to attain gender inequality in all the enlisted spheres. We believe our use of gender equality in education relates more to these goals.

The rest of the paper proceeds as follows: Sections 2 and 3 present the literature review and methodology of the study, respectively. Section 4 presents and discusses the empirical results, and Section 5 presents the concluding remarks.

2. Literature review

2.1. Direct linkage between energy poverty and gender equality in education

The concept and definition of energy poverty are as complex, inconclusive, and inexhaustible as economic/income poverty. As a result, many scholars have explained the concept from many viewpoints; for example, energy poverty has been defined as having no access to the socially and materially essential domestic energy services (for example, heating, lighting, or cooking) that enable one to meaningfully participate in the society (Robinson, 2019). Some regard it as the lack of adequate choice in accessing suitable, affordable, reliable, high-quality, and environmentally safe energy services to support day-to-day activities, whether social or economic (Jayasinghe et al., 2021) or the level of modern energy consumption that is inadequate to meet some fundamental needs of people (González-Eguino, 2015). Generally and concisely, energy poverty is the lack of or inadequate access to modern, sustainable, and affordable energy (Longe, 2021; Ngarava et al., 2022). In the last couple of decades, there has been a tremendous positive trend in access to energy, mainly due to the decentralization of energy generation and distribution to reach rural and remote areas (Pearl-Martinez, 2020) and several programs run and championed by the World Bank, a number of the United Nations' organizations, governments, and other international and local organizations. For example, in SSA (Sub-Saharan Africa), access to electricity has increased from about 26 % of the population in 2000 to 33 % in 2010 and 48 % in 2020 (see Fig. 1). Globally, access to electricity has increased from about 78 % of the population in 2000 to 83 % in 2010 and 91 % in 2020 (see Fig. 1).

Though energy poverty affects everyone, women, especially in developing countries, are believed to suffer most as they are household energy's main users and producers (Bouzarovski and Petrova, 2015; Pearl-Martinez, 2020; Sánchez-Guevara Sánchez et al., 2020; Ssennono et al., 2021). In their study, Ngarava et al. (2022) found that women are more energy vulnerable in South Africa, where energy vulnerability captures individuals' inability to secure the energy service required at home. Ssennono et al. (2021) similarly find that women are the most energy-poor and vulnerable in Uganda. Aristondo and Onaindia (2018) find that women are 10 % poorer in energy than men in Spain on average. Hence, gender energy inequality must be addressed to achieve just, equitable, sustainable, and modern energy. Robinson (2019) identifies five dimensions of gendered energy vulnerability- i.e., what makes women more energy vulnerable; time-consuming and unpaid activities, caring or performance of domestic roles; exposure to physiological and mental health impacts; a lack of social protection during a life course; and coping and helping others to cope.

Interestingly, time is disproportionately allocated between men and women in meeting household energy demands (Ozughalu and Ogwumike, 2018; Rewald, 2017; Ssennono et al., 2021). In many developing countries where households lack access to modern energy, using traditional biomass for cooking and meeting other household energy demands is common. The arduous and time-consuming nature of searching, collecting, and managing traditional biomass is largely viewed as women's and girls' duty (Kaygusuz, 2011; Ssennono et al., 2021). Longe (2021) asserts that in many developing countries, women and girls are commonly stereotyped with the task of providing and using unclean energy for their households. Robinson (2019) asserts that the performance of domestic and caring duties is usually considered responsibilities of women due to socially-constructed gender norms. He argues that these dimensions usually make women more energy vulnerable. Chaudhry and Shafiullah (2021), in their work (involving 103 countries), show how culture affects energy poverty. Abdourahman (2010) argues that in many patriarchal cultural settings, women, relative to men, are overburdened with tasks in the household, one of the major tasks being the collection of traditional biomasses. Collecting these unclean energies or biomasses could take considerable time due to the long walking distances involved.

From the discussion above, it can be conjectured that the lack or inadequate access to modern energy would dispropotionaly affect girls and women in many aspects of their lives including education. Long distances walked by girls and women to collect firewood for example will directly reduce the time they have to spent at and on school. The IEA (2020a) estimates that on the average about 5 h are spent a day collecting fuel for cooking in rural parts of some developing countries; Sierra Leone, 4 h in Niger, 3.1 h in Senegal, 1.7 h in Sudan and Nigeria, 1.6 h in Ghana, and 1.1 h in India. The share of the population that relies on traditional biomass for cooking in these countries is 97.4 %, 95.8 %, 67.6 %, 64.8 %, 70 %, 71.2 % and 59.5 % respectively (IEA, 2020a). Girls and women may spend a significant amount of time performing household chores using traditional energy sources, which can reduce the time they have available for attending school. Access to modern energy sources can help reduce the time spent on household chores and allow for more time to be devoted to education. The stress they have to go through in search of this energy and cooking with it can also reduce interest and time spent in school. Although boys and men may experience similar challenges, the gender gap in energy access and usage disproportionately affects women, making them more vulnerable to the negative impacts on their education. In many developing countries, particularly the poor ones where energy poverty is prevalent, boys tend to have higher enrollment and completion rates in schools compared to girls. For example, in low-income countries, girls' primary (78 %) and secondary school (31 %) enrollment rates are lower than global averages of 88 % and 66 %, respectively, while in fragile, conflict, and violence-prone countries, girls are more likely to drop out of school (Kattan and Khan, 2023). Sub-Saharan Africa has the highest gender disparities in education, with about 86 million of girls out of school, and low-middle income and Sub-Saharan African countries have very low tertiary enrollment rates for girls compared to the global average. The completion rates for lower secondary school are also low for girls in low-income (38 %) and Sub-Saharan African countries (43 %) (Kattan and Khan, 2023).

2.2. Transmission channels through which energy poverty affects gender equality in education

In the proceeding sections, we discuss some possible channels through which energy poverty or access to energy could contribute to gender equality in education.

2.2.1. Employment, savings, and financial security

A major social and economic value that comes with the transition to modern energy (access to sustainable and modern energy) is time-

[1] https://www.un.org/sustainabledevelopment/development-agenda/

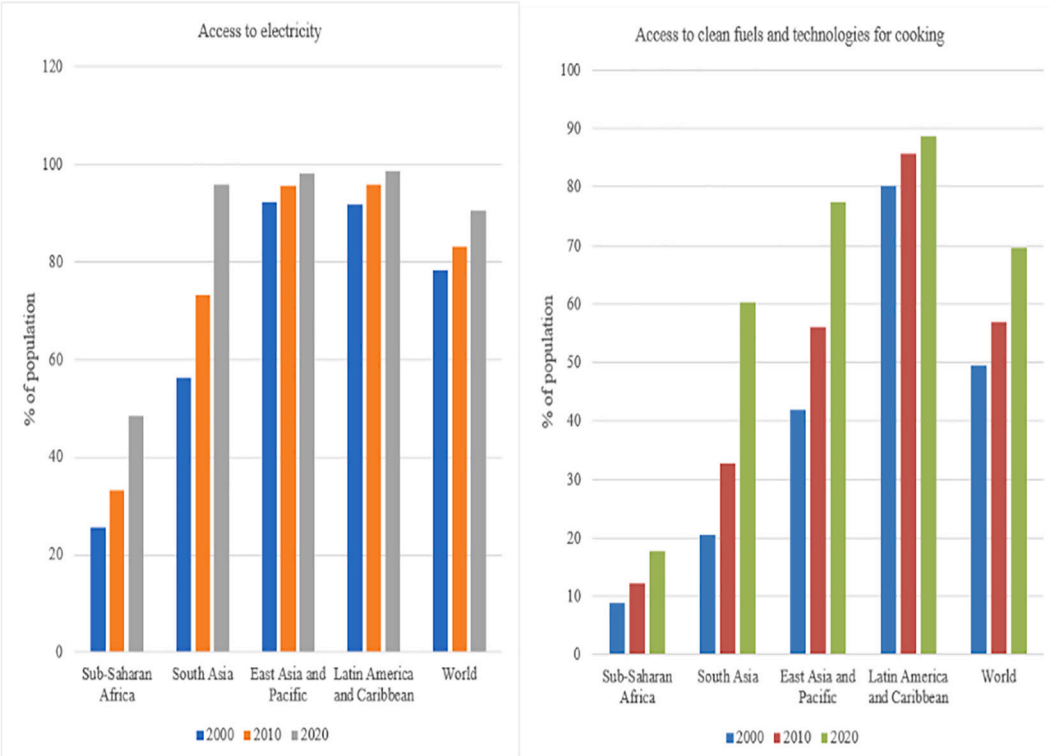

Fig. 1. Access to electricity and clean fuels and technologies for cooking. NB: SSA, SA, EAP, and LAC represent Sub-Saharan Africa, South Asia, East Asia and Pacific, and Latin America and the Caribbean.
Source: Authors', with data from the World Bank Development Indicators.

saving. Transition and access to clean and modern energy would reduce the search time for energy, and release more time for other important activities like paid jobs (Opoku et al., 2021; Ssennono et al., 2021). The transition and access to modern and clean energy could save households money, increasing the chances of girls being sent to school (Pueyo and Maestre, 2019). In many poor households in developing countries, males are chosen over females regarding education. One of the reasons has been financial challenges. However, with access to modern and clean energy, the release of time from biomass energy search and collection can i) make people work longer and earn more wages from paid jobs. Time released from other household activities could give extra time for households to engage in more income-earning activities that can make them earn more (Ozughalu and Ogwumike, 2018; Ssennono et al., 2021). Ding et al. (2014) show how integrating renewable energy increased the saving of time and money for women in north-west China. Working enhances financial inclusion, and as Koomson and Danquah (2021) note, financial inclusion could likely lead to the most significant decrease in energy poverty. With reduced energy poverty, households tend to enjoy the benefits that come with energy access ii) reduce health bills due to reduced or "zero" pollution from clean and modern energy, iii) enhance refrigeration/preservation of food that drastically reduces wastage which indirectly leads to money saving (Puri, 2016). iv) Women can work and contribute to household income (Pueyo and Maestre, 2019).

Dinkelman (2011) shows that providing electricity in rural South Africa increased women's employment by about 9 %. Grogan and Sadanand (2009) also found a 23 % likelihood for women to work outside the home with access to electricity in Nicaragua. Barron and Torero (2014) find that in Northern El Salvador, electrification led to higher participation in income-generating activities among women. In a study of 51 developing countries from 2002 to 2017, Nguyen and Su (2021) found that reducing energy poverty increases women's employment opportunities. A study by Rewald (2017) finds that numerous modern energy access projects did not meet the intended purpose for females because i) households continued to use traditional energy sources because they are relatively cheaper. Firewood, for example, is the most affordable alternative to clean energy in many developing countries (UN, 2021). Some biomass typically has no direct monetary cost (Rewald, 2017). ii) females are less involved in decision-making, so they cannot decide between working home chores or working outside the home. Fingleton-Smith (2018), in a study, shows that men do not spend much time at home but carry out major household purchasing decisions, including energy. This paper argues that it is due to the traditional gender power divisions. Winther et al. (2020) found similar results for Nepal, Kenya, and India.

2.3. Health

Another importance of access to modern and clean energy is that it improves the health of women and girls (Jayasinghe et al., 2021; Nguyen and Su, 2021; Pachauri and Rao, 2013; Pueyo and Maestre, 2019). Nguyen and Su (2021) found evidence from a sample of 51 countries that access to modern and clean energy improves women's health. Using a global dataset, Pan et al. (2021) found that energy poverty is detrimental to public health, and the effect is greater for women. Women, girls, and children face significant sicknesses and premature death from the dangers of collecting fuels and indoor air pollution (Abbas et al., 2020; Pearl-Martinez, 2020). The inefficient burning of biomass generates a mix of harmful pollutants, including carbon monoxide, nitrogen oxides, benzene, butadiene, and many other poisonous chemicals (Sovacool, 2012). These chemicals can lead to stroke, ischaemic heart disease, chronic obstructive pulmonary disease, eye problems, lung cancer, cardiovascular disease, asthma, and respiratory infections (Sovacool, 2012; WHO, 2021). Household air pollution causes about 4 million premature deaths annually (WHO, 2021). It is estimated that about 60 % of these deaths are women and children who

mostly clung to their mothers when cooking (Rewald, 2017). Globally, the WHO considers household air pollution to be the second most significant health risk factor (WHO, 2021). Women and girls also face physical injury risks that harm their health in collecting biomass (Rewald, 2017; Sovacool, 2012). Longe (2021), in his study, revealed that energy poverty (inaccessibility of electricity) in some communities of selected provinces in South Africa exposes women and girls to security issues and health dangers, among other disadvantages. It has also been found that burning solid biomass at homes without a chimney lowers children's IQ, and this affects their education (Ssennono et al., 2021). Energy poverty also impedes effective medical care in communities that lack electricity. The health effect of energy poverty can affect women's and girls' education through lateness to school, increased absenteeism, complete dropout and inability to concentrate on school activities due to illness (Ozughalu and Ogwumike, 2018; Sovacool, 2012). It was found that a third of school absenteeism in some rural areas in Uganda was as a result of acute respiratory infections (ACI) from indoor air pollution. Each episode of the ACI lasted 7–9 days for each child.

2.4. Information and communication technology

Access to modern energy also enhances education through television, the use of computers, the internet, and other modern media (Pachauri and Rao, 2013; Rewald, 2017). With television, computers, and the internet, students (in this case, girls and women) can acquire information and learn from various sources across the globe. A study in rural India revealed that women with access to cable television were likelier to send their young girls to school (Jensen and Oster, 2009). Cobb (2004), and, Agarwal and Day (1998), and many other studies have shown how internet technology enhances education. During the recent Covid-19 global lockdowns, internet-connected computers became an essential educational alternative platform for students worldwide. A study by Young Lives showed that in Vietnam, online classes were prevalent during the pandemic, with a larger percentage of participation by female students (89 %) relative to males (86 %) (Scott et al., 2020). The work of EdTech Hub shows that access to technology empowers girls and women more than boys and men (Webb et al., 2020). The EdTech Hub also shows that with access to technology and technology-enabled education, girls respond with a relatively higher engagement level (Webb et al., 2020). iMlango (an e-learning program) has supported about 70,130 marginalized girls in developing their learning and movement to the next level of education (iMlango, 2021).

2.5. Literacy and academic performance

Access to electricity can also prolong the time children have to learn and enable schools to be equipped with electrical learning devices (like computers) (Sovacool, 2012). Energy services can also help schools attract, recruit, and retain quality teachers (Cabraal et al., 2005; Sovacool, 2012). Khandker et al. (2014) find that children in rural India with electricity at home learn 2.3 h more at home relative to those without electricity access. They also find that access to electricity increases enrolment and retention at school. Barron and Torero (2014) find that in Northern El Salvador, access to electricity increases the likelihood for children to study at home and perform school-connected activities by about 54 % and 84 %, respectively. In Mali, Brew-Hammond and Crole-Rees (2004) indicated that the introduction of electricity improved girls' school attendance and academic performance. In Bangladesh, Barkat et al. (2002) found that school attendance increased with the introduction of electric fans. They further find that literacy is 20 % higher among women with household electricity than those without. In their study, Barnes and Sen (2004) found that about 90 % of women who read in India are those with electricity at home. However, Bensch et al. (2011) did not find any significant effect of access to electricity on children's home learning in Rwanda.

Based on the extensive literature review, the following hypotheses are formulated:

H1. Access to energy directly increases gender equality in education.

H2. Female employment mediates the effect of energy access on gender equality in education.

H3. Female health mediates the effect of energy access on gender equality in education.

H4. Information and communication technology mediate the effect of energy access on gender equality in education.

H5. Female literacy mediates the effect of energy access on gender equality in education.

3. Methodology and data

3.1. Specification of empirical models

Our main empirical model for estimating the impact of access to energy on gender equality in education takes the form:

$$GED_{i,t} = \alpha_0 + \beta_1 ENERGY_{i,t} + \theta_1 + ln\boldsymbol{X}_{i,t} + \varepsilon_{it} \quad (1)$$

where $GED_{i,t}$ is gender equality in education of country i at year t. $ENERGY_{i,t}$ is access to energy in the country i at year t. $ln\boldsymbol{X}_{i,t}$ vector of control covariates that affect gender equality in education. ε_{it} is the unobserved error term. a_0, β_1 and θ_1 are unknown coefficients to be estimated.

3.1.1. Models for examining the potential pathways

Based on the conceptualization in the literature section, we test if female literacy, female health, female employment, and ICT serve as the transmission mechanisms through which access to energy can affect gender equality in education. In doing so, we perform pathway (mediation) analysis to validate if the variables indeed mediate the effect of access to energy on gender equality in education. In so doing, two-staged approaches consistent with practice (Acheampong et al., 2021a; Acheampong et al., 2021b; Awaworyi Churchill and Smyth, 2020) are followed. In the first stage, for these variables to serve as mediators through which access to energy influences gender equality in education, they must be correlated with access to energy variables. We model the first stage assumption on the relationship between access to energy variables and the selected mediating variables using Eq. (2).

$$PMV_{i,t} = \alpha_1 + \varnothing_1 ENERGY_{i,t} + \varnothing_2 lnX_{i,t} + \varepsilon_{i,t} \quad (2)$$

where $PMV_{i,t}$ represents the potential mediating variables (female literacy, female health, female employment, and ICT) i at year t. After the first stage, we proceed to the second stage. In the second stage, the access to energy variables found to have a statistically significant relationship with female literacy, female health, female employment, and ICT would be included as an additional explanatory variable in the regression linking access to energy to gender equality in education (Eq. (3)). Supposing female literacy, health, employment, and ICT are full mediators; the assumption is that their inclusion as additional explanatory variables in the gender equality in education model should decrease the magnitude of the coefficient on access to energy variables or render them statistically insignificant.

$$GED_{i,t} = \alpha_2 + \delta_1 ENERGY_{i,t} + \delta_2 \mathrm{PMV}_{i,t} + \theta_3 ln\boldsymbol{X}_{i,t} + \varepsilon_{it} \quad (3)$$

a_1, a_2, $\varnothing_1$, $\varnothing_2$, δ_1 and δ_2 in Eqs. (2) and (3) are unknown coefficients to be estimated.

3.2. Estimation techniques

In this study, we apply the Baum et al. (2002) instrumental variable generalized method of moments (two-step GMM). The two-step GMM

technique helps to cater for endogeneity, which is one of the significant challenges in applied research.[2] In addition to catering for the endogeneity issue, the two-step GMM estimator allows consistent estimations in the presence of autocorrelation (AR, (1)) within panels and heteroscedasticity (Baum et al., 2002). Also, the two-step GMM estimator is consistent with Driscoll and Kraay (1998) standard errors that are robust to 'spatial' and temporal cross-sectional dependence even when the time dimension becomes relatively large. The two-step GMM technique has been applied in a number of applied research (for instance, Acheampong et al., 2021a; Acheampong et al., 2023; Acheampong et al., 2022; Boateng et al., 2019; Boateng et al., 2021). We also deploy the Driscoll and Kraay (1998) estimator to test the robustness of the two-step GMM findings. The Driscoll and Kraay (1998) estimator assumes that the error structure of Eq. (1) is heteroscedastic, autocorrelated up to some lag, and perhaps correlated between the panels. This suggests that the Driscoll and Kraay (1998) estimator is essential for capturing autocorrelation and heteroskedasticity. Also, the Driscoll–Kraay estimator generates robust estimates of both cross-sectional and temporal dependence (Hoechle, 2007). The Driscoll and Kraay (1998) estimator can also handle missing data series and works with balanced and unbalanced panels (Hoechle, 2007). We further test the consistency of the findings using the Feasible Generalized Least Squares (FGLS). FGLS also yields efficient estimates in the presence of heteroskedasticity and serial and cross-sectional correlations (Bai et al., 2021).

3.3. Description and sources

We used panel data from 98 countries between 2000 and 2021 in this study.[3] These 98 countries were sampled from SSA, South Asia, Latin America & Caribbean regions.

3.3.1. Dependent variable

Following Nguyen and Su (2021) and Cooray and Potrafke (2011), three (3) variables were used to represent gender equality in education: (1) Gender equality in primary education measured with gender parity index for primary school gross enrollment; (2) Gender equality in secondary education measured with gender parity index for secondary school gross enrollment; (3) Gender equality in tertiary education measured with gender parity index for tertiary school gross enrollment. All these variables were sourced from the World Development Indicators (WDI) of the World Bank database.

3.3.2. Independent variables

Following Acheampong et al. (2022), Acheampong et al. (2021b) and Nguyen and Su (2021), we measured energy poverty with six (6) variables: (1) Access to clean cooking fuels and technologies measured with the percentage of the population with access to clean fuels and technologies for cooking; (2) Rural access to clean cooking fuels and technologies measured with the percentage of rural population with access to clean fuels and technologies for cooking; (3) Urban access to clean cooking fuels and technologies measured with the percentage of urban population with access to clean fuels and technologies for cooking; (4) Electrification measured with the percentage of population with access to electricity; (5) Rural electrification measured with the percentage of rural population with access to electricity; (6) Urban electrification measured with the percentage of urban population with access to electricity. These variables were obtained from WDI. Following from the discussion in the literature review, we expect positive coefficients for the variables measuring energy poverty, hence their improvement increases gender parity in education (Karlsson and McDade, 2001; Lambrou and Piana, 2006; Nguyen and Su, 2021).

3.3.3. Control variables

Following the literature on determinants of gender inequality, standard variables such as GDP per capita (economic growth), access to credit, trade openness, foreign direct investment, education expenditure, internet penetration, and foreign aid were used as the control variables (Brown, 2004; Chen et al., 2013; Cooray and Potrafke, 2011; Nguyen and Su, 2021). Economic growth is measured with GDP per capita (constant 2015 US$). GDP per capita (economic growth) as a measure of average purchasing power and income level of individuals is expected to have a positive coefficient on gender parity in education (Cooray and Potrafke, 2011; Nguyen and Su, 2021). The implication is that with an increase in economic growth, households and governments will be able to invest in education to bridge the gender gap. Trade openness is proxied with trade (export + import) (% of GDP), and foreign direct investment is measured with net inflows of foreign direct investment (% of GDP). The effect of trade openness on the dependent variable could be ambiguous; on the one hand, it could worsen gender parity in education because, in many developing countries, unskilled women workers have been employed in labour-intensive export industries, hence taking them out of school (Cooray and Potrafke, 2011; Fontana and Wood, 2000). On the other hand, trade openness is expected to bridge the wage gap between skilled and unskilled workers, and gender. With higher relative wages, women may be able to access greater educational opportunities, hence increasing gender parity in education (Cooray and Potrafke, 2011). Foreign aid is measured with net official development assistance and official aid received (constant 2020 US$). Aid could also have ambiguous effect on the dependent variable; the effect could be positive, negative or insignificant (Maïga, 2014; Pickbourn and Ndikumana, 2016). Foreign aid, if directed to what it is meant for, e.g. to address gender inequality issues, could have a positive effect on the gender parity of education. Nonetheless, aid can sometimes be redirected and create opportunities for corruption, which can ultimately have little to no impact or even a negative effect on educational and other outcomes (Maïga, 2014). Education expenditure is measured with total government expenditure on education as a percentage of GDP. It is expected to have a positive effect on the dependent variable, as an increase in education expenditure would increase access and opportunities for both boys and girls in education. Access to credit is measured with domestic credit to private sector by banks as a percentage of GDP. Access to credit is expected to have a positive effect; it provides an avenue for households to acquire credit to fund their children's education and to expand their businesses to have the financial means to see their children (including girls) through school. Internet penetration is measured with individuals using the internet as a percentage of population. Greater internet penetration is expected to have positive effect on the dependent variable. Higher internet penetration and usage enhance education and enable education beyond the brick and mortar education (Nguyen and Su, 2021). As a result, internet can increase education for all including girls.

3.3.4. Mediating variables

Based on the literature review, four variables were considered for the mediation analysis: (1) Female literacy measured with youth female literacy rate as a percentage of females ages 15–24; (2) Female health measured with female life expectancy at birth; (3) Female employment

[2] There is a consensus that finding a true exogenous instrument is often challenging task for applied research using instrumental variable estimators (Stock et al., 2002). In view of this, we used lag one and two of access to energy variables (electricity and clean cooking technologies) as instruments to cater for endogeneity. Several reasons motivate this study to use the lags of energy access variables as the instrument. For instance, policies and investments to facilitate access to electricity and clean cooking technologies take time to materialize and impact outcomes. Therefore, the lags of energy access variables are important for capturing the time it takes for energy policies to be implemented to impact population access to energy. Also, in developing countries, it takes time for communities and people to connect to the national and off-grid; therefore, using the lags of energy access variables is important for isolating the exogenous effect of energy access on gender equality in education.

[3] See the Appendix Table 1 for countries included in the study.

measured with female employers as a percentage of female employment; (4) ICT captured using ICT goods, telephone, and mobile phone penetration. ICT goods are measured with ICT service exports as a percentage of service exports. Telephone penetration is proxied with fixed telephone subscriptions, and mobile phone penetration is proxied with mobile cellular subscriptions. These variables were obtained from the WDI. Following from the literature, we expect these variables to have positive coefficients; hence, their improvement would enhance gender parity in education (Dinkelman, 2011; Nguyen and Su, 2021; Pueyo and Maestre, 2019; Scott et al., 2020).

These control variables were obtained from the WDI database. The summary statistics for the variables are presented in Appendix Table 2. For estimation, all the variables were log-transformed. Figs. 2A to 2F and 3 show that access to energy variables positively correlates with gender equality in primary, secondary, and tertiary education.

4. Empirical findings

We present the estimates on the impact of access to energy and the control variables on gender equality in education in Table 1. From Table 1, Columns 1–6 show the results for using the gender parity index for primary education as the dependent variable, while Columns 7–12 display the results for using the gender parity index for secondary education as the dependent variable. Columns 13–18 present the results using the gender parity index for tertiary education as the dependent variable.

Table 1 shows that access to clean cooking fuels and technologies has a statistically insignificant effect on the gender parity index for primary [see Column 1] but has a statistically significant positive effect on the gender parity index for secondary and tertiary education at a 1 % level [see Columns 7 and 13]. The coefficients show that a 1 % increase in clean cooking fuels and technologies is associated with a 0.024 and 0.052 percentage point increase in gender parity in secondary and tertiary education, respectively, ceteris paribus. It can be observed that rural access to clean cooking fuels and technologies has a stronger positive association with the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 2, 8, and 14]. The estimated coefficients show that a 1 % increase in rural access to clean cooking fuels and technologies is associated with 0.013, 0.048, and 0.106 percentage point increases in gender parity in primary, secondary, and tertiary education, respectively, ceteris paribus. The estimated coefficients show that while urban access to clean cooking fuels and technologies has a statistically insignificant effect on the gender parity index for primary education [see Column 3], it has a statistically significant effect on the gender parity index for secondary and secondary education a 1 % level [see Columns 9 and 15]. Thus, a 1 % increase in urban access to clean cooking fuels and technologies is associated with an increase in the gender parity index for secondary and tertiary education by 0.022 and 0.048 percentage points, respectively, ceteris paribus.

Further evidence from Table 1 shows that the estimated coefficient of electrification has a stronger positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 4, 10, and 16]. The estimated coefficient shows that a 1 % increase in electrification improves the gender parity index for primary, secondary, and tertiary education by 0.033, 0.084, and 0.119 percentage points, respectively, ceteris paribus. Similarly, rural electrification has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 5, 11, and 17]. The estimated coefficient shows that a 1 % increase in rural electrification improves the gender parity index for primary, secondary, and tertiary education by 0.015, 0.040, and 0.061 percentage points, respectively, ceteris paribus. It is observed that urban electrification also has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 6, 12, and 18]. The estimated coefficients indicate that a 1 % increase in urban electrification improves the gender parity index for primary, secondary, and tertiary education by 0.105, 0.283, and 0.333 percentage points, respectively, ceteris paribus.

Generally, the results support that increasing access to energy could ensure equality in education between females and males. In developing countries, women are vulnerable to energy poverty since they are the main users of energy (Pearl-Martinez, 2020; Sánchez-Guevara Sánchez et al., 2020; Ssennono et al., 2021). In developing countries with a lack of access to clean cooking fuels, women and girls spend a significant proportion of their time collecting and gathering traditional biomass for cooking and meeting other household energy demands (Ozughalu and Ogwumike, 2018; Rewald, 2017; Ssennono et al., 2021). The arduous and time-consuming nature of searching, collecting, and managing traditional biomass is largely viewed as women's and girls' duty (Kaygusuz, 2011; Ssennono et al., 2021), and this can hinder their enrolment in schools. Backing our results with Amartya Sen's Capability to Development theory, our findings, therefore, indicate that improving energy access could enable vulnerable people, especially women and girls, to enhance their capability to function effectively (Acheampong et al., 2021a). Thus, with increasing access to energy, gender equality in education could be enhanced since access to energy can enable women to build and own assets. Energy access can improve gender equality in education by improving employment, income, academic performance, literacy and health. Access to clean and modern energy could enhance gender equality in education by saving women and girls a significant amount of time dedicated to the collection of biomasses for cooking and channelling that time towards productive and academic activities. Also, evidence has shown that increasing access to modern and clean energy could save households money and increase the chances of girls being sent to school (Pueyo and Maestre, 2019).

Next, we turn to the control variables. Generally, the estimates show that access to credit, a proxy for financial development, has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education, suggesting that increasing access to credit is associated with an increase in gender equality in education. Financial constraints have prevented many people (including girls) from accessing and completing education at all levels of education (Blimpo et al., 2016). The literature has shown that in the face of financial challenges, parents give less premium to girls' education and therefore, girls have less tendency to enroll in education relative to boys (Lloyd and Young, 2009; Psaki et al., 2022). Therefore, with access to credit, parents could give equal attention to their girls' education and invest in their girls' access to and completion of school.

It is observed that while GDP per capita mostly has a statistically significant negative effect on the gender parity index for primary and secondary education, it has a statistically significant positive effect on the gender parity index for tertiary education. This observation shows that economic growth widens gender inequality in education at the lower levels of the education ladder [primary and secondary education) while improving gender equality in education at tertiary education. Relative to primary and secondary education, the cost of attending tertiary education is higher. The higher cost of attending tertiary education has been a major driver of widening gender disparity in tertiary education. Our results, therefore, reflect that to increase women's enrolment in tertiary education, resources generated by higher economic growth are used to finance educational programmes, including the provision of scholarships that would benefit women as well. Arora (2012) equally documented that higher GDP per capita could be accompanied by either increasing or reducing gender equality in education. However, our results are inconsistent with the findings of Cooray and Potrafke (2011) and Maïga (2014) that GDP per capita has an insignificant effect on gender parity in primary and secondary education.

It is observed that, generally, foreign aid has a statistically insignificant effect on the gender parity index for primary and secondary education but a statistically significant negative effect on the gender parity

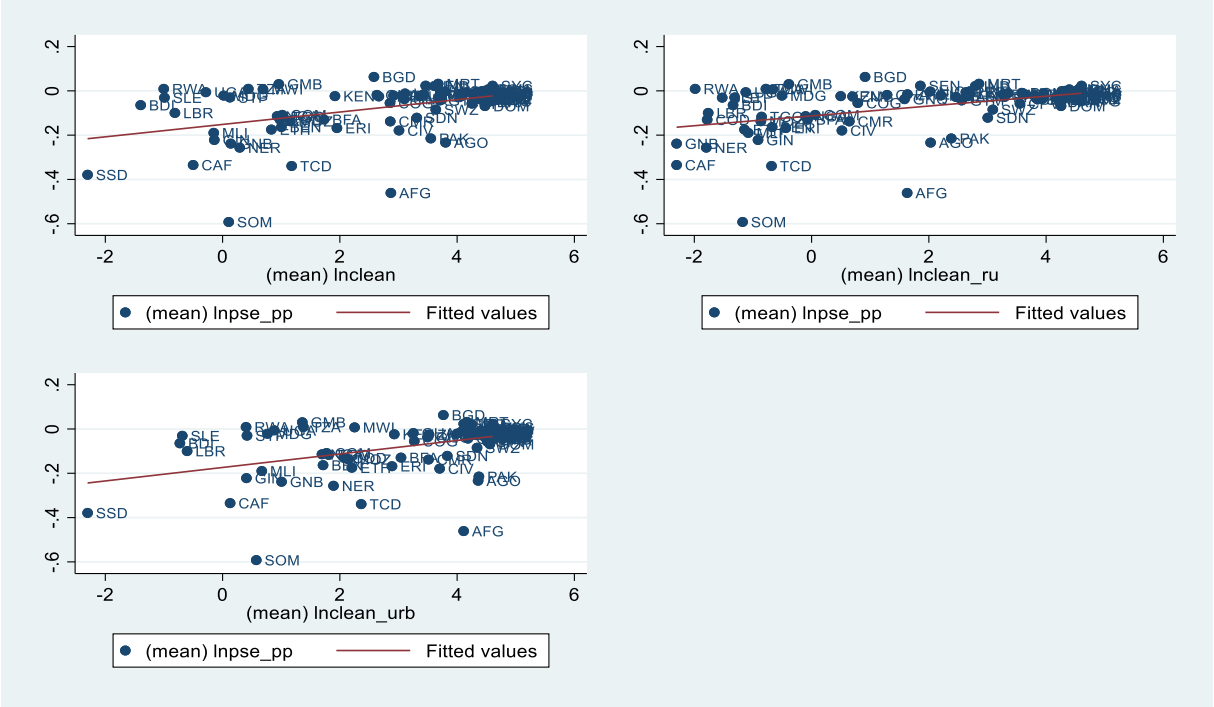

Fig. 2A. Bivariate correlation between clean cooking fuels and technologies variables and gender equality in primary education.

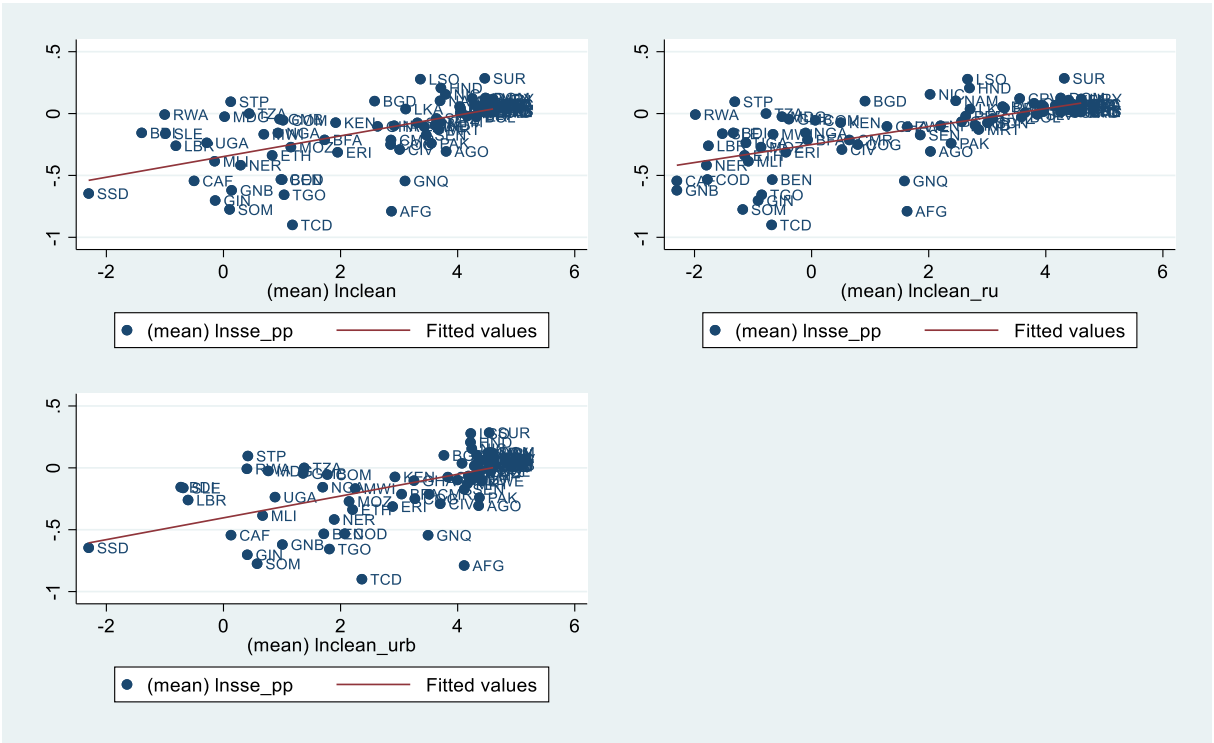

Fig. 2B. Bivariate correlation between clean cooking fuels and technologies variables and gender equality in secondary education.

index for tertiary education. This result indicates that foreign aid is associated with widening gender disparity in tertiary education. The observed effect of foreign aid on gender equality in education could reflect "aid fungibility". Thus, in developing countries, foreign aid that is disbursed to address gender inequality issues in education is re-allocate by recipient country's governments to other sectors of this economy, thereby making foreign aid ineffective in impacting gender equality in primary and secondary education and to larger extent worsen gender inequality in tertiary education. These findings align with the evidence presented by (Maïga, 2014) that foreign aid significantly reduces gender parity in tertiary education but has no effect on gender parity in primary education.

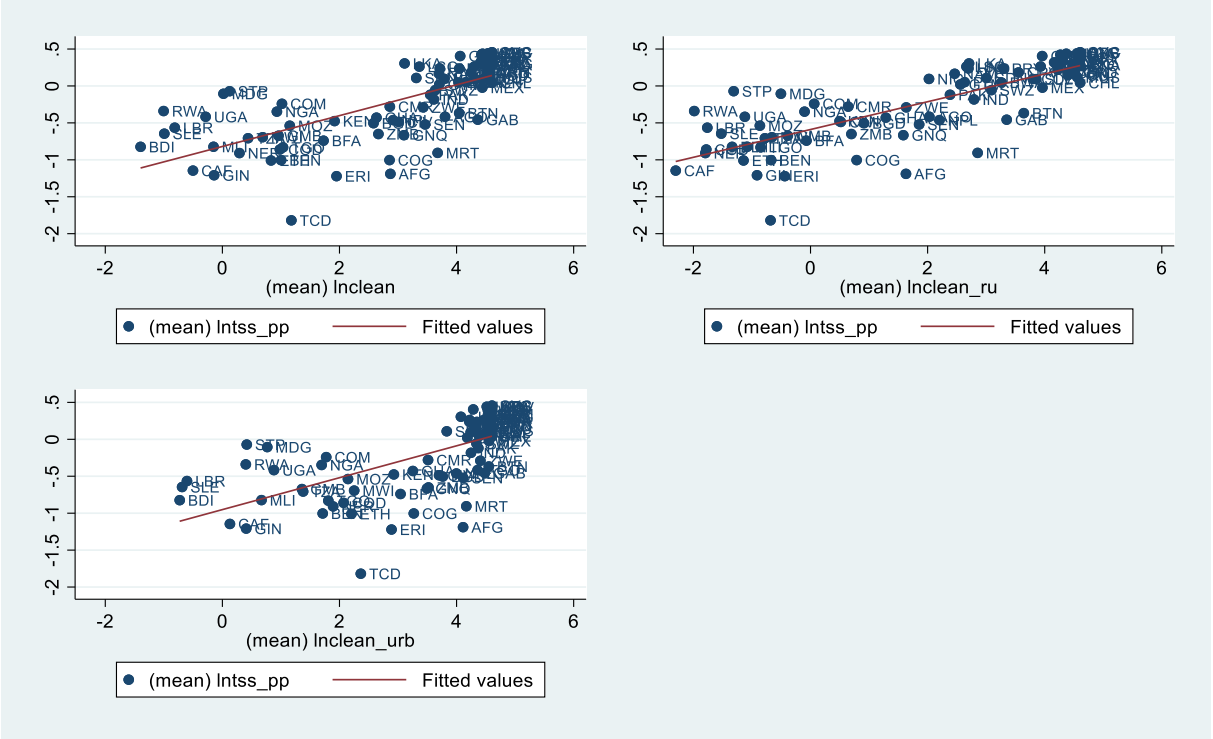

Fig. 2C. Bivariate correlation between clean cooking fuels and technologies variables and gender equality in tertiary education.

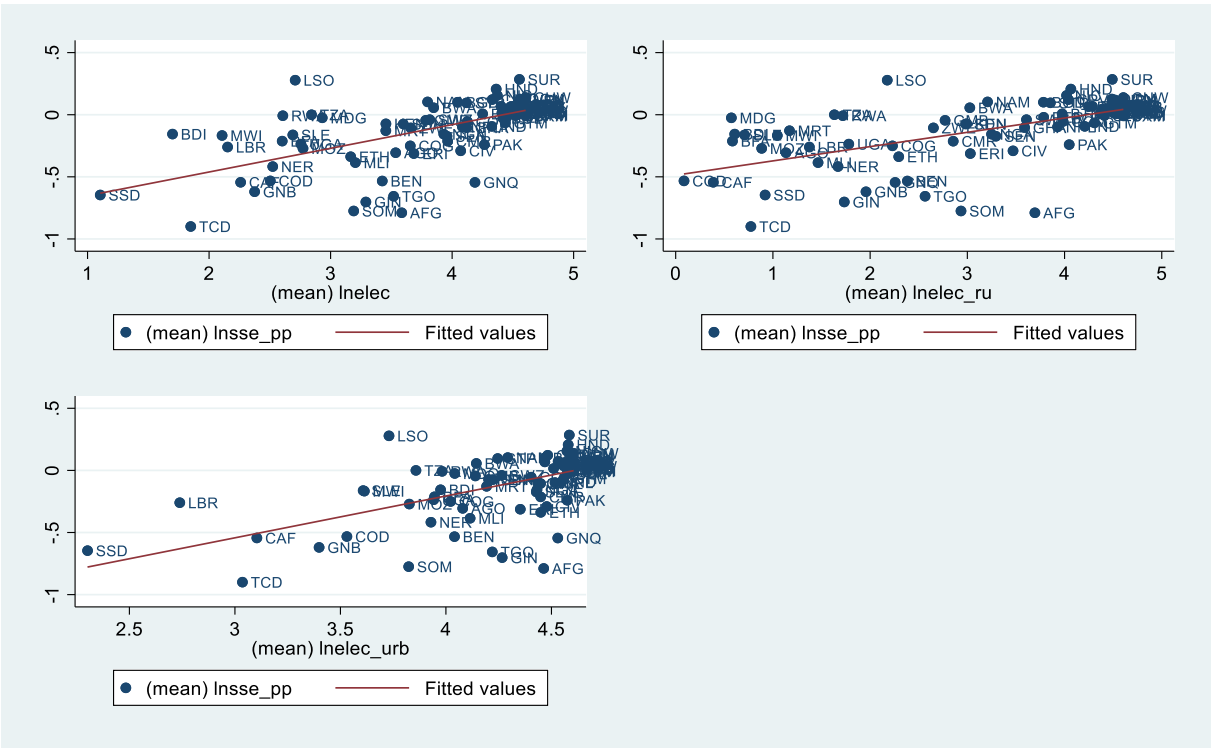

Fig. 2D. Bivariate correlation between access to electricity variables and gender equality in primary education.

Inconsistent with the findings of Lloyd and Young (2009), our results indicate that foreign direct investment mostly has a statistically significant negative effect on the gender parity index for primary and secondary education but has a statistically insignificant positive effect on the gender parity index for tertiary education. The implication is that increasing inflows of foreign direct investment in developing regions widen gender inequality in education. In other words, multinational corporations in developing countries bring managerial and technical skills that favour men at the expense of women, thereby enlarging gender inequality in education. Also, foreign direct investment could

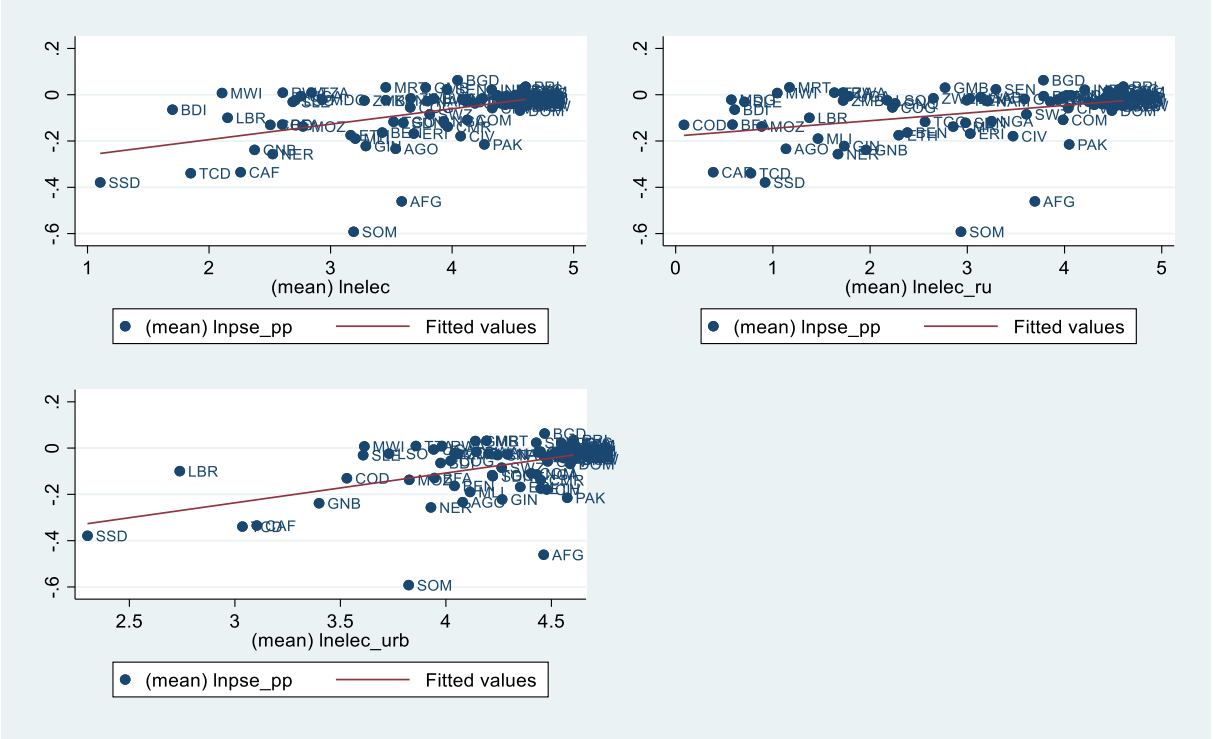

Fig. 2E. Bivariate correlation between access to electricity variables and gender equality in secondary education.

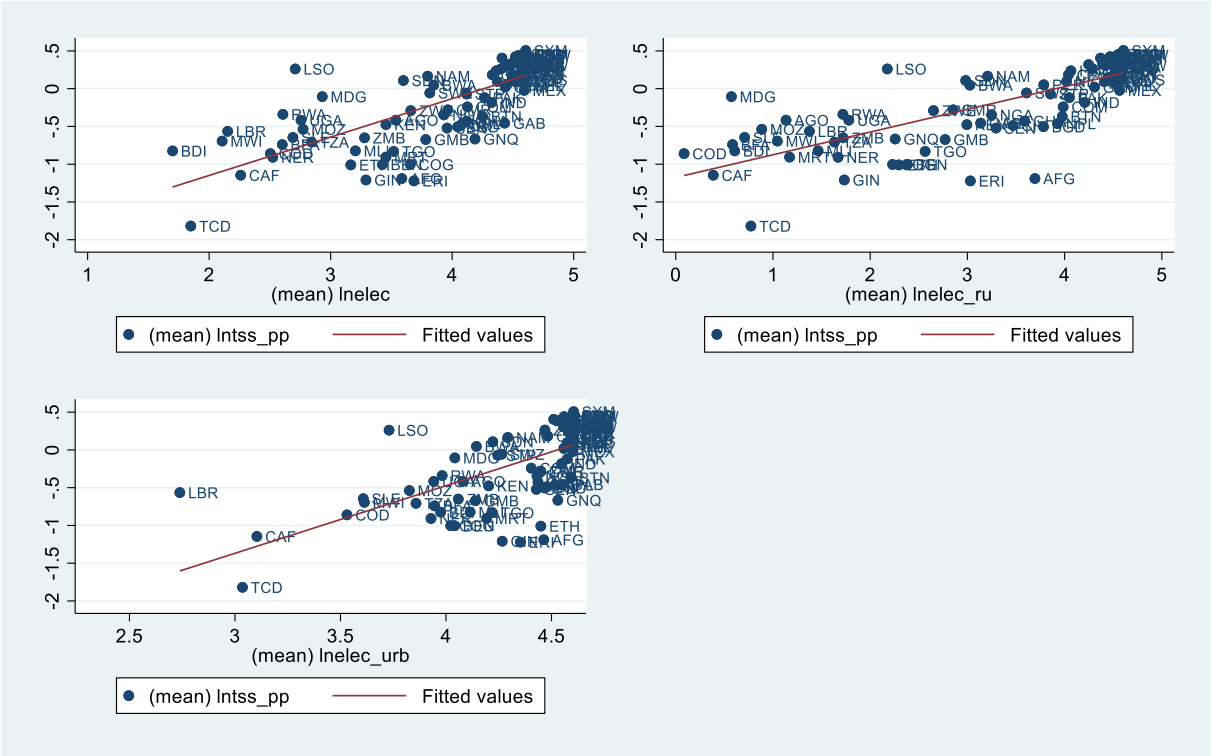

Fig. 2F. Bivariate correlation between access to electricity variables and gender equality in tertiary education.

reduce gender equality in education when it leads to the creation of job opportunities that favour males, and this can distort parents' investment in girl-child education.

Trade openness has a statistically significant positive effect on the gender parity index for primary while it has a statistically insignificant effect on the gender parity index for secondary and tertiary education. With reference to Korinek (2005), trade openness could enhance gender equality, especially in education, since trade creates jobs and provides a lot of opportunities in the export-oriented sectors, which generates financial resources that could lead to greater investment in girls'

Table 1
Energy poverty and gender equality in education [two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to credit	0.015***	0.009*	0.014***	0.013***	0.013**	0.008*	0.058***	0.035***	0.059***	0.054***	0.047***	0.034***	0.065**	0.017	0.070***	0.063***	0.060**	0.052**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.012)	(0.013)	(0.012)	(0.010)	(0.011)	(0.010)	(0.026)	(0.027)	(0.026)	(0.024)	(0.026)	(0.024)
GDP per capita	−0.008	−0.033***	−0.004	−0.024***	−0.017***	−0.024***	0.007	−0.038**	0.013	−0.001	0.018	0.004	0.103***	0.000	0.125***	0.116***	0.144***	0.136***
	(0.007)	(0.007)	(0.006)	(0.006)	(0.007)	(0.005)	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.013)	(0.032)	(0.035)	(0.029)	(0.031)	(0.027)	(0.028)
Foreign aid	0.005	−0.000	0.007**	0.001	0.001	−0.000	−0.004	−0.001	−0.006	−0.005	0.000	−0.002	−0.030**	−0.024*	−0.029**	−0.027**	−0.016	−0.026**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.007)	(0.006)	(0.006)	(0.007)	(0.007)	(0.006)	(0.013)	(0.012)	(0.013)	(0.014)	(0.012)	(0.013)
Foreign direct investment	−0.005*	−0.001	−0.006**	−0.002	0.001	−0.001	−0.010*	−0.004	−0.010*	−0.009	−0.005	−0.010*	0.003	0.017	0.002	0.002	0.010	0.009
	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.006)	(0.006)	(0.006)	(0.006)	(0.008)	(0.005)	(0.010)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Trade openness	0.023**	0.005	0.030***	0.016*	0.009	0.017**	−0.013	−0.023	−0.013	−0.006	0.002	0.022	−0.043	−0.055	−0.043	−0.021	0.003	−0.018
	(0.009)	(0.008)	(0.009)	(0.009)	(0.009)	(0.008)	(0.019)	(0.018)	(0.019)	(0.020)	(0.021)	(0.018)	(0.037)	(0.036)	(0.037)	(0.038)	(0.036)	(0.036)
Education expenditure	0.015	−0.003	0.014	0.014	0.008	0.001	0.054**	0.037*	0.054**	0.050**	0.034	0.025	0.023	0.035	0.009	0.009	−0.063	−0.011
	(0.011)	(0.010)	(0.011)	(0.011)	(0.012)	(0.010)	(0.024)	(0.022)	(0.024)	(0.022)	(0.026)	(0.022)	(0.062)	(0.064)	(0.061)	(0.062)	(0.066)	(0.062)
Internet penetration	0.034***	0.032***	0.035***	0.029***	0.026***	0.026***	0.049***	0.052***	0.051***	0.042***	0.032***	0.038***	0.103***	0.106***	0.103***	0.095***	0.087***	0.084***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.020)	(0.019)	(0.020)	(0.021)	(0.021)	(0.021)
Access to clean cooking fuels and technologies	0.000						0.024***						0.052***					
	(0.004)						(0.009)						(0.016)					
Access to clean cooking fuels and technologies (rural)		0.013***						0.048***						0.106***				
		(0.004)						(0.009)						(0.018)				
Access to clean cooking fuels and technologies (urban)			−0.002						0.022**						0.048***			
			(0.004)						(0.009)						(0.016)			
Electrification				0.033***						0.084***						0.119***		
				(0.010)						(0.021)						(0.037)		
Rural electrification					0.015**						0.040***						0.061***	
					(0.007)						(0.013)						(0.022)	
Urban electrification						0.105***						0.283***						0.333***
						(0.015)						(0.040)						(0.082)
Constant	−0.349***	0.077	−0.426***	−0.228**	−0.144	−0.482***	−0.521**	−0.097	−0.529**	−0.674***	−0.669**	−1.650***	−0.888*	−0.133	−1.060**	−1.382***	−1.443***	−2.461***
	(0.127)	(0.107)	(0.124)	(0.110)	(0.115)	(0.109)	(0.241)	(0.225)	(0.236)	(0.254)	(0.263)	(0.244)	(0.513)	(0.530)	(0.489)	(0.481)	(0.454)	(0.458)
Observations	663	652	663	665	598	658	560	555	560	562	509	557	504	499	504	505	458	502
R^2	0.327	0.307	0.327	0.353	0.296	0.392	0.482	0.501	0.480	0.504	0.434	0.549	0.628	0.648	0.627	0.631	0.641	0.646
Hansen	2.699	1.259	14.799	2.190	0.148	0.208	0.006	1.520	2.047	2.476	0.159	0.075	0.046	0.621	1.784	0.488	0.478	0.015
Hansen (*p*-value)	0.100	0.262	0.000	0.139	0.700	0.648	0.938	0.218	0.153	0.116	0.690	0.784	0.831	0.431	0.182	0.485	0.489	0.903
F-statistics	31,138.493	20,846.991	47,368.551	3909.303	624.711	755.688	24,639.938	18,269.342	36,444.762	3113.908	551.515	764.920	19,098.855	13,939.343	31,139.424	2942.370	554.233	368.535
No. of instruments	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

Table 2
Energy poverty and Gender equality in Education [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to credit	0.007	0.006	0.006	0.010*	0.010*	0.004	0.026**	0.018	0.023*	0.029**	0.027**	0.010	0.012	−0.003	0.011	0.014	0.052*	0.003
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)	(0.011)	(0.030)	(0.029)	(0.030)	(0.030)	(0.031)	(0.030)
GDP per capita	−0.031***	−0.051***	−0.027***	−0.047***	−0.039***	−0.044***	−0.058***	−0.086***	−0.055***	−0.065***	−0.047**	−0.054***	0.016	−0.038	0.028	0.036	0.099***	0.050
	(0.007)	(0.008)	(0.007)	(0.007)	(0.007)	(0.006)	(0.017)	(0.017)	(0.016)	(0.019)	(0.020)	(0.016)	(0.037)	(0.038)	(0.035)	(0.037)	(0.035)	(0.034)
Foreign aid	0.004	−0.001	0.005*	−0.000	−0.001	−0.001	−0.003	0.001	−0.006	−0.005	0.001	−0.002	−0.021	−0.018	−0.024	−0.018	−0.010	−0.018
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.008)	(0.007)	(0.007)	(0.009)	(0.009)	(0.008)	(0.016)	(0.015)	(0.016)	(0.016)	(0.015)	(0.015)
Foreign direct investment	−0.003	0.000	−0.004	0.001	0.003	0.002	−0.004	0.002	−0.002	0.000	0.003	−0.000	0.012	0.022**	0.012	0.011	0.012	0.017*
	(0.003)	(0.002)	(0.003)	(0.003)	(0.004)	(0.003)	(0.006)	(0.006)	(0.006)	(0.007)	(0.009)	(0.005)	(0.010)	(0.010)	(0.010)	(0.010)	(0.011)	(0.010)
Trade openness	0.019*	−0.006	0.025***	0.006	−0.002	0.006	−0.037*	−0.054***	−0.037*	−0.033	−0.030	−0.005	−0.043	−0.068*	−0.044	−0.021	−0.002	−0.020
	(0.010)	(0.009)	(0.009)	(0.010)	(0.010)	(0.008)	(0.019)	(0.018)	(0.019)	(0.021)	(0.021)	(0.017)	(0.042)	(0.041)	(0.041)	(0.041)	(0.039)	(0.038)
Education expenditure	0.017	−0.007	0.017	0.011	0.005	−0.005	0.030	0.003	0.034	0.023	0.008	−0.004	0.007	−0.011	0.006	−0.007	−0.112*	−0.041
	(0.011)	(0.009)	(0.011)	(0.011)	(0.012)	(0.010)	(0.023)	(0.021)	(0.024)	(0.023)	(0.026)	(0.022)	(0.061)	(0.063)	(0.060)	(0.061)	(0.060)	(0.058)
Internet penetration	0.031***	0.028***	0.031***	0.024***	0.021***	0.021***	0.036***	0.038***	0.035***	0.026***	0.017**	0.021***	0.083***	0.084***	0.086***	0.075***	0.072***	0.063***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.019)	(0.019)	(0.019)	(0.020)	(0.021)	(0.020)
Age dependency ratio	−0.002***	−0.002***	−0.002***	−0.002***	−0.002***	−0.002***	−0.007***	−0.006***	−0.007***	−0.007***	−0.007***	−0.006***	−0.010***	−0.008***	−0.010***	−0.010***	−0.005***	−0.009***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
CPI	0.004***	0.003***	0.004***	0.003***	0.002*	0.002***	0.006***	0.005***	0.006***	0.004**	0.003	0.004**	0.009**	0.006	0.009**	0.008*	0.009**	0.006
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Sex ratio at birth	−0.974***	−1.516***	−0.832**	−1.464***	−1.267***	−1.493***	−3.734***	−4.475***	−3.698***	−4.002***	−4.162***	−3.937***	−2.530**	−3.809***	−2.324**	−2.548**	−2.237*	−2.714***
	(0.345)	(0.361)	(0.340)	(0.348)	(0.370)	(0.333)	(0.607)	(0.629)	(0.602)	(0.619)	(0.661)	(0.481)	(1.139)	(1.172)	(1.121)	(1.234)	(1.157)	(1.040)
Access to clean cooking fuels and technologies	0.002						0.025***						0.047***					
	(0.004)						(0.008)						(0.017)					
Access to clean cooking fuels and technologies (rural)		0.013***						0.049***						0.083***				
		(0.004)						(0.010)						(0.022)				
Access to clean cooking fuels and technologies (urban)			−0.001						0.023***						0.047***			
			(0.004)						(0.009)						(0.016)			
Electrification				0.041***						0.092***						0.094**		
				(0.010)						(0.023)						(0.045)		
Rural electrification					0.018***						0.040**						0.051*	
					(0.007)						(0.016)						(0.027)	
Urban electrification						0.117***						0.294***						0.330***
						(0.015)						(0.033)						(0.078)
Constant	1.034**	2.000***	0.811*	1.653***	1.560***	1.420***	4.610***	5.553***	4.629***	4.677***	4.908***	3.561***	3.165**	4.884***	2.864*	2.670*	1.687	1.776
	(0.426)	(0.429)	(0.416)	(0.415)	(0.446)	(0.384)	(0.805)	(0.796)	(0.788)	(0.850)	(0.923)	(0.658)	(1.546)	(1.533)	(1.493)	(1.556)	(1.446)	(1.263)
Observations	632	621	632	634	571	627	529	524	529	531	482	526	474	469	474	475	431	472
R2	0.396	0.384	0.395	0.415	0.361	0.458	0.576	0.587	0.574	0.588	0.540	0.637	0.651	0.656	0.650	0.651	0.649	0.667
Hansen	5.368	0.381	17.417	0.609	0.031	0.020	0.241	0.757	2.884	1.310	0.010	0.001	0.414	0.053	2.110	0.080	0.532	0.000

(*continued on next page*)

Table 2 (*continued*)

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Hansen (p-value	0.021	0.537	0.000	0.435	0.860	0.886	0.623	0.384	0.089	0.252	0.920	0.979	0.520	0.819	0.146	0.778	0.466	0.995
F-statistics	29,388.434	12,641.305	42,895.809	3264.764	393.590	629.464	21,976.315	9331.284	31,510.848	2402.825	377.032	631.145	16,376.191	8662.917	27,733.897	2263.769	397.636	283.573
Number of instruments	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
Number of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

education. This result corroborates the findings of Ben Saâd and Assoumou-Ella (2019) and Korinek (2005) that trade openness or trade sophistication improves gender equality in education.

Internet penetration has a statistically significant positive effect on the gender parity index for primary, secondary and tertiary education. This result indicates that improving access to the internet is key to attaining gender equality in education. This finding is not far from the literature that access to the internet and ICT in general has a significant positive net effect for girls/women relative to boys/men. For instance, a study by Scott et al. (2020) indicated that in Vietnam, online classes were prevalent during the pandemic, with a larger percentage of participation by female students (89 %) relative to males (86 %). Similarly, a study conducted by EdTech Hub shows that access to the internet, and technology empowers girls and women more than boys and men (Webb et al., 2020). The EdTech Hub also shows that with access to technology and technology-enabled education, girls respond with a relatively higher engagement level (Webb et al., 2020). The implication of increasing access to the internet is that girls/women could have access to educational resources that could improve their literacy and academic performance.

Education expenditure is observed to have a statistically significant positive effect on the gender parity index for secondary education and not gender parity index for primary and tertiary education. This result indicates that government spending on education is only important for enhancing gender equality in secondary school and not primary and tertiary education. It is not surprising that educational expenditure has an insignificant effect on gender equality in primary education. In developing countries, there has been significant access to primary education by all genders; however, there has been a significant gap between girls and boys in secondary education, which is attributed to financial constraints (Blimpo et al., 2016). Therefore, government allocation of finance to support educational programmes and interventions aimed at enhancing girls' access to secondary education is vital for addressing gender inequality in education. Educational programs and interventions in the form of scholarships would eliminate the financial barriers that hinder girls from enrolling and completing school. Our evidence that educational expenditure has a significant positive effect on gender parity in school education confirms the findings of Ben Saâd and Assoumou-Ella (2019).

4.1. Additional control variables

In this section, we test the effects of energy access on gender equality in education by including additional control variables in our models. Age dependency ratio, consumer price index (CPI) and sex ratio at birth are the additional variables controlled in the empirical models.[4] The results presented in Table 2 are not qualitatively different from the results presented in Table 1. In other words, the inclusion of age dependency ratio, consumer price index, and sex ratio at birth does not change the impact of energy access variables on gender equality in education. For instance, the results presented in Table 2 confirm Table 1 results by indicating that while access to clean cooking fuels and technologies has a statistically insignificant effect on gender parity index for primary education, it has a statistically significant positive effect on the gender parity index for secondary and tertiary education. Also, rural access to clean cooking fuels and technologies has a stronger positive association with the gender parity index for primary, secondary, and tertiary education. In addition, the results in Table 2 also confirm that while urban access to clean cooking fuels and technologies has a

[4] These varables were suggested by reviewers. Age dependency ratio is measured with age dependency ratio (% of working-age population); sex ratio at birth, a proxy for social norms, measured with sex ratio at birth (male births per female births) and consumer price index (CPI), measured as consumer price index (2010 = 100).

Table 3
The effect of energy poverty on female literacy, health, and employment [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Female literacy						Female health						Female employment					
Access to clean cooking fuels and technologies	0.004*						0.012***						−0.136**					
	(0.023)						(0.004)						(0.055)					
Access to clean cooking fuels and technologies (rural)		0.025						0.019***						−0.013				
		(0.021)						(0.004)						(0.050)				
Access to clean cooking fuels and technologies (urban)			0.002						0.011***						−0.266***			
			(0.024)						(0.004)						(0.044)			
Electrification				0.154**						0.061***						−0.036		
				(0.062)						(0.008)						(0.098)		
Rural electrification					0.125***						0.043***						−0.029	
					(0.038)						(0.005)						(0.064)	
Urban electrification						0.418***						0.141***						−0.373**
						(0.103)						(0.016)						(0.148)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	2.688***	3.478***	2.574***	2.793***	3.084***	1.546***	4.169***	4.471***	4.142***	4.064***	4.221***	3.741***	−5.799***	−3.874***	−7.483***	−4.029***	−5.576***	−3.494***
	(0.516)	(0.461)	(0.502)	(0.482)	(0.457)	(0.593)	(0.124)	(0.116)	(0.120)	(0.106)	(0.105)	(0.092)	(1.192)	(1.178)	(1.117)	(0.990)	(1.149)	(1.007)
Observations	251	249	251	251	231	248	804	787	804	807	716	797	791	774	791	793	702	783
R^2	0.603	0.607	0.603	0.636	0.639	0.658	0.679	0.682	0.678	0.698	0.687	0.726	0.305	0.285	0.332	0.293	0.326	0.294
Hansen	2.026	3.651	0.024	2.961	0.249	0.175	7.715	0.689	12.222	0.166	1.351	0.332	2.523	1.353	5.051	1.180	0.479	2.456
Hansen (p-value)	0.155	0.056	0.878	0.085	0.618	0.676	0.005	0.407	0.000	0.684	0.245	0.565	0.112	0.245	0.025	0.277	0.489	0.117
F-statistics	13,726.868	4439.128	9467.192	795.451	124.312	1272.082	38,303.305	30,773.338	41,196.443	3097.213	850.466	1142.942	38,040.751	30,346.261	41,691.515	3088.443	840.474	1143.915
No. of instruments	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

Table 4
The effect of energy poverty on ICT [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Mobile cellular subscriptions						ICT goods						Fixed telephone subscriptions					
Access to clean cooking fuels and technologies	0.070						**0.195*****						**0.586*****					
	(0.045)						**(0.052)**						**(0.048)**					
Access to clean cooking fuels and technologies (rural)		−0.052						**0.051**						**0.511*****				
		(0.046)						**(0.049)**						**(0.044)**				
Access to clean cooking fuels and technologies (urban)			**0.079***						**0.191*****						**0.492*****			
			(0.042)						**(0.058)**						**(0.046)**			
Electrification				**0.230****						**0.680*****						**1.127*****		
				(0.091)						**(0.110)**						**(0.118)**		
Rural electrification					**0.129****						**0.489*****						**0.687*****	
					(0.062)						**(0.061)**						**(0.072)**	
Urban electrification						**0.352****						**0.810*****						**1.927*****
						(0.152)						**(0.272)**						**(0.256)**
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−2.037	−3.060**	−2.204	−2.747**	−1.568	−3.214**	8.879***	6.712***	8.692***	5.656***	7.110***	3.546**	1.814	2.473	0.427	−5.584***	−2.596	−10.976***
	(1.560)	(1.525)	(1.532)	(1.367)	(1.480)	(1.362)	(1.485)	(1.431)	(1.468)	(1.311)	(1.195)	(1.765)	(1.901)	(1.691)	(1.877)	(1.749)	(1.767)	(1.758)
Observations	803	786	803	806	714	793	737	736	737	740	668	727	784	770	784	785	700	772
R^2	0.728	0.723	0.729	0.730	0.722	0.725	0.118	0.100	0.121	0.144	0.210	0.113	0.737	0.734	0.729	0.725	0.728	0.724
Hansen	0.439	0.004	0.178	2.485	2.554	0.070	0.132	0.873	1.582	1.570	0.071	0.171	5.199	2.214	8.040	0.496	2.642	0.909
Hansen (p-value)	0.508	0.949	0.673	0.115	0.110	0.792	0.716	0.350	0.209	0.210	0.789	0.680	0.023	0.137	0.005	0.481	0.104	0.340
F-statistics	40,012.257	31,707.215	51,645.542	3110.510	837.713	1122.367	29,945.198	28,117.535	32,176.087	2148.181	820.949	629.054	34,938.366	25,646.043	37,878.370	3654.566	831.699	1246.877
No. of instruments	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

Table 5
Analysis of potential channels: Energy poverty, female literacy, and Gender equality in Education [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9
	Gender Parity Index for Primary School			Gender Parity Index for Secondary School			Gender Parity Index for Tertiary School		
Female literacy	0.175***	0.157***	0.159***	0.469***	0.528***	0.439***	0.517***	0.463***	0.512***
	(0.020)	(0.027)	(0.022)	(0.047)	(0.060)	(0.053)	(0.106)	(0.126)	(0.106)
Electrification	**−0.018**			**0.004**			**0.124****		
	(0.016)			**(0.036)**			**(0.054)**		
Rural electrification		**−0.014***			**−0.030****			**0.037**	
		(0.010)			**(0.014)**			**(0.032)**	
Urban electrification			**0.018**			**0.117**			**0.208**
			(0.038)			**(0.089)**			**(0.134)**
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.766***	−0.690***	−0.726***	−2.123***	−2.513***	−2.412***	−2.592***	−2.617***	−3.497***
	(0.154)	(0.184)	(0.143)	(0.372)	(0.406)	(0.391)	(0.615)	(0.708)	(0.643)
Observations	205	189	202	187	173	184	172	160	171
R^2	0.488	0.408	0.493	0.722	0.695	0.732	0.726	0.676	0.726
Hansen	2.712	1.288	0.242	0.256	0.104	0.196	1.102	0.846	2.203
Hansen (p-value)	0.100	0.256	0.622	0.613	0.747	0.658	0.294	0.358	0.138
F-statistics	573.368	94.864	940.942	469.132	85.926	996.687	458.578	72.760	1397.120
No. of instruments	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98
Panel B [Initial results for Comparison, see Table 2]									
Electrification	**0.033*****			**0.084*****			**0.119*****		
	(0.010)			**(0.021)**			**(0.037)**		
Rural electrification		**0.015****			**0.040*****			**0.061*****	
		(0.007)			**(0.013)**			**(0.022)**	
Urban electrification			**0.105*****			**0.283*****			**0.333*****
			(0.015)			**(0.040)**			**(0.082)**

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

statistically insignificant effect on the gender parity index for primary education, it has a statistically significant effect on the gender parity index for secondary and secondary education. Also, observation from the results in Table 2 highlights that electrification has a stronger positive effect on the gender parity index for primary, secondary, and tertiary education. Consistently, the rural electrification variable also has a positive and statistically significant effect on the gender parity index for primary, secondary, and tertiary education. In the same manner, the urban electrification variable has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education.

The estimates presented in Table 2 indicate that the age dependency ratio has a statistically significant and negative effect on the gender parity index for primary, secondary, and tertiary education. This finding implies that increasing age dependency is detrimental to attaining gender equality in education at all levels. Similarly, the sex-at-birth ratio variable has a statistically significant and negative effect on the gender parity index for primary, secondary, and tertiary education. By construction, increasing sex at birth ratio indicates more births given to males relative to females. This result, therefore, implies that giving more birth to males worsens gender inequality in education across all educational levels. It is observed from Table 2 that the consumer price index (CPI) has a statistically significant positive effect on gender parity index for primary, secondary and tertiary education.

4.2. Analysis of the transmission channels

In this section, we perform a mediation analysis to identify potential channels through which access to energy influences gender equality in education. In conducting the mediation analysis, we follow the existing literature to use the conventional procedure, which involves a two-stage approach to test if female literacy, female health, female employment, and ICT (ICT goods, mobile phone, and telephone penetration) are the potential channels through which access to energy affect gender equality in education. We test the first condition first and then proceed to test the second condition to validate if these variables mediate the relationship between access to energy and gender equality in education.

4.2.1. Testing for the first stage condition

The first stage condition assumes that access to energy variables should have a significant relationship with potential mediators (female literacy, female health, female employment, mobile phone subscription, ICT goods, telephone penetration). To verify this, we present the statistical relationship between the access to energy variables and potential mediators (in Tables 3 and 4. Table 3 shows that access to clean cooking fuels and technologies and rural and urban access to clean cooking fuels and technologies have statistically insignificant effects on female literacy [Columns 1–3]. The estimated coefficient shows that electrification and rural and urban electrification have a statistically significant positive effect on female literacy [see Columns 4–6]. These findings suggest that in testing the second stage condition, female literacy will not be included as an explanatory variable in the models linking access to clean energy for cooking variables to gender equality in education. However, female literacy will be included as an independent covariate in the models linking electrification, rural electrification, and urban electrification variables to gender equality in education.

In Table 3, Columns 7–12 suggest that access to clean cooking fuels and technologies, rural access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification have a statistically significant positive effect on female health at a 1 % level. These findings suggest that in testing the second stage condition, female health will be included as an explanatory variable in the models linking clean cooking fuels and technologies, rural access to clean cooking fuels and

Table 6
Analysis of potential channels: Energy poverty, female health, and Gender equality in Education [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Female literacy						Female health						Female employment					
Female health	0.359***	0.292***	0.360***	0.335***	0.287***	0.286***	0.463***	0.360***	0.468***	0.426***	0.348***	0.257***	0.965***	0.736***	0.922***	0.858***	0.452**	0.696***
	(0.035)	(0.033)	(0.035)	(0.037)	(0.038)	(0.038)	(0.100)	(0.098)	(0.099)	(0.096)	(0.100)	(0.098)	(0.197)	(0.202)	(0.198)	(0.207)	(0.219)	(0.233)
Access to clean cooking fuels and technologies	−0.004						0.018*						0.053***					
	(0.004)						(0.009)						(0.016)					
Access to clean cooking fuels and technologies (rural)		0.007*						0.040***						0.096***				
		(0.004)						(0.010)						(0.018)				
Access to clean cooking fuels and technologies (urban)			−0.007**						0.015*						0.048***			
			(0.004)						(0.009)						(0.016)			
Electrification				0.014						0.056***						0.073**		
				(0.010)						(0.022)						(0.037)		
Rural electrification					0.003						0.023						0.046**	
					(0.006)						(0.014)						(0.023)	
Urban electrification						0.061***						0.240***						0.229**
						(0.018)						(0.045)						(0.098)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−1.861***	−1.238***	−1.948***	−1.615***	−1.378***	−1.562***	−2.661***	−1.807***	−2.684***	−2.562***	−2.255***	−2.642***	−4.751***	−3.392***	−4.774***	−4.943***	−3.335***	−5.038***
	(0.191)	(0.187)	(0.189)	(0.196)	(0.214)	(0.168)	(0.558)	(0.548)	(0.541)	(0.502)	(0.548)	(0.439)	(0.972)	(1.064)	(0.967)	(0.985)	(1.093)	(0.973)
Observations	661	650	661	663	598	658	559	554	559	561	509	557	504	499	504	505	458	502
R^2	0.443	0.387	0.444	0.442	0.369	0.456	0.516	0.522	0.515	0.525	0.452	0.558	0.651	0.661	0.649	0.647	0.646	0.656
Hansen	1.165	2.135	8.685	2.871	0.480	0.402	0.251	2.184	0.704	3.221	0.334	0.099	0.425	0.471	1.131	0.682	0.703	0.021
Hansen (p-value)	0.281	0.144	0.003	0.090	0.488	0.526	0.616	0.139	0.401	0.073	0.563	0.753	0.514	0.493	0.287	0.409	0.402	0.885
F-statistics	29,241.491	18,765.891	44,782.292	3286.516	506.396	678.894	22,043.143	16,804.665	33,488.869	2582.960	451.555	612.857	19,267.255	12,917.304	31,108.252	2513.257	492.990	376.924
No. of instruments	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98
Panel B [Initial results for Comparison, see Table 2]																		
Access to clean cooking fuels and technologies	0.000						0.024***						0.052***					
	(0.004)						(0.009)						(0.016)					
Access to clean cooking fuels and technologies (rural)		0.013***						0.048***						0.106***				
		(0.004)						(0.009)						(0.018)				

(*continued on next page*)

Table 6 (*continued*)

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Female literacy						Female health						Female employment					
Access to clean cooking fuels and technologies (urban)			−0.002						0.022**						0.048***			
			(0.004)						(0.009)						(0.016)			
Electrification				0.033***						0.084***						0.119***		
				(0.010)						(0.021)						(0.037)		
Rural electrification					0.015**						0.040***						0.061***	
					(0.007)						(0.013)						(0.022)	
Urban electrification						0.105***						0.283***						0.333***
						(0.015)						(0.040)						(0.082)

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification to gender equality in education. Also, Table 3, Columns 13–18 shows that access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, and urban electrification have a statistically significant negative effect on female employment. However, rural access to clean cooking fuels and technologies, electrification, and rural electrification have a statistically insignificant effect on female employment. The implication is that the female employment variable will be included as an independent variable in the models linking access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, and urban electrification to gender equality in education.

In Table 4, Columns 1 and 2 show that access to clean cooking fuels and technologies and rural access to clean cooking fuels and technologies have no significant effect on mobile phone penetration. However, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification have a statistically significant positive effect on mobile phones [see Table 4, Columns 3–6]. This result shows that in the second stage analysis, mobile phone penetration will be included as an independent variable in the models linking urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification to gender equality in education. Also, in Table 4, Columns 7–12 show that except for rural access to clean cooking fuels and technologies variable, access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification have a statistically significant positive effect on ICT goods at a 1 % level. In addition, access to clean cooking fuels and technologies, rural access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification have a statistically significant positive effect on telephones at a 1 % level [see Table 4, Columns 13–18]. These results imply that in the second stage analysis, ICT goods and telephone phone penetration will be included as independent variables in the models linking access to energy variables to gender equality in education.

4.2.2. 4.2.2. Testing for the second stage condition

The variables that passed the statistical significance test in the first stage analysis are used to test the second stage condition. The second stage condition assumes that if female literacy, female health, female employment, mobile phone subscription, ICT goods, and telephone penetration are mediators, their inclusion in the gender equality in education models should either render the access to energy variables statistically insignificant or reduce the magnitude of the coefficients. To ascertain this assumption, the coefficients and statistical significance of the access to energy variables in Panel A of Tables 5–10 should be compared to their respective coefficients and statistical significance in Panel B of these Tables, which are drawn from Table 1.

In Table 5, female literacy has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education, suggesting that female literacy contributes significantly to enhancing gender equality at all levels of education. In Panel A, it is observed that the inclusion of female literacy in the models renders electrification, rural electrification, and urban electrification to have a statistically insignificant effect on the gender parity index for primary education. Also, the inclusion of female literacy as an independent variable renders the impact of electrification and urban electrification on the gender parity index for secondary education statistically insignificant while reducing the size of rural electrification coefficient from 0.040 to 0.030, ceteris paribus. Finally, except for the electrification variable, rural and urban electrification have a statistically insignificant effect on the gender parity index for tertiary education when female literacy is included as an independent variable in the models. These findings show that female literacy is one of the potential channels through which access to electricity and rural and urban electrification affect gender equality in education.

Table 7
Analysis of potential channels: energy poverty, female employment, and gender equality in education [two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9
	Gender Parity Index for Primary School			Gender Parity Index for Secondary School			Gender Parity Index for Tertiary School		
Female employment	0.011***	0.011***	0.011***	0.018**	0.020**	0.019***	0.081***	0.085***	0.079***
	(0.003)	(0.003)	(0.003)	(0.008)	(0.008)	(0.007)	(0.015)	(0.015)	(0.014)
Access to clean cooking fuels and technologies	0.001*			0.026***			0.062***		
	(0.004)			(0.009)			(0.018)		
Access to clean cooking fuels and technologies (Urban)		−0.001			0.025***			0.062***	
		(0.004)			(0.009)			(0.016)	
Urban electrification			0.105***			0.286***			0.344***
			(0.015)			(0.039)			(0.075)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.231*	−0.293**	−0.412***	−0.285	−0.237	−1.463***	−0.202	−0.252	−1.956***
	(0.129)	(0.126)	(0.111)	(0.265)	(0.258)	(0.260)	(0.547)	(0.520)	(0.478)
Observations	649	649	645	547	547	544	496	496	493
R^2	0.338	0.337	0.402	0.487	0.487	0.552	0.643	0.644	0.658
Hansen	1.460	11.771	0.450	0.040	1.283	0.234	0.586	0.056	0.118
Hansen (p-value)	0.227	0.001	0.502	0.841	0.257	0.628	0.444	0.813	0.731
F-statistics	29,338.935	49,302.186	758.578	24,767.101	39,041.585	773.365	19,096.214	32,159.365	365.292
No. of instruments	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98
Panel B [initial results for comparison, see Table 2]									
Access to clean cooking fuels and technologies	**0.000**			**0.024*** **			**0.052*** **		
	(0.004)			**(0.009)**			**(0.016)**		
Access to clean cooking fuels and technologies (Urban)		**−0.002**			**0.022** **			**0.048*** **	
		(0.004)			**(0.009)**			**(0.016)**	
Urban electrification			**0.105*** **			**0.283*** **			**0.333*** **
			(0.015)			**(0.040)**			**(0.082)**

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

It is observed in Table 6 that female health has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education, indicating that enhancing female health contributes significantly to addressing gender inequality at all levels of education. Panel A of Table 6 suggests that except for access to clean cooking fuels and technologies and urban access to clean cooking fuels and technologies, the magnitude of rural access to clean cooking fuels and technologies and urban electrification coefficients on the gender parity index in primary education reduced. For instance, the coefficient of rural access to clean cooking fuels and technologies reduced from 0.013 to 0.007, ceteris paribus, while the coefficient of urban electrification reduced from 0.105 to 0.061, ceteris paribus. In addition, the impact of electrification and rural electrification on the gender parity index in primary education becomes statistically insignificant after including the female health variable in the models. Also, comparing the coefficients in Panel A to the coefficients in Panel B, it is observed that the size of the impact of access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification on the gender parity index in secondary education is reduced when female health is included as an additional covariate in the models. Also, except for access to clean cooking fuels and technologies and urban access to clean cooking fuels and technologies, the size of the impact of urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification on the gender parity index in tertiary education is reduced when female health is included as an additional covariate in the models [comparing the coefficients in Panel A to the coefficients Panel B]. These findings generally indicate that female health is one of the potential pathways through which access to electricity and clean cooking fuels and technologies affect gender equality in education.

It is observed from Table 7 that female employment has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education, indicating that self-employed females contribute substantially to improving gender inequality at all levels of education. In Table 7, Panel A reveals that the impact (coefficient) of urban access to clean cooking fuels and technologies on the gender parity index for primary education reduced after including female employment as an additional covariate in the model. For instance, the coefficient of urban access to clean cooking fuels and technologies reduced from 0.002 [see Panel B] to 0.001 [see Panel A], ceteris paribus. However, the inclusion of female employment as an independent variable does not either reduce the coefficients or render the impact of access to clean cooking fuels and technologies and urban electrification on the gender parity index for primary education statistically insignificant. Also, it can be observed that the inclusion of female employment as an independent variable does not either reduce the coefficients or render the impact of access to clean cooking fuels and technologies, rural access to clean cooking fuels and technologies and urban electrification on gender parity index for secondary and tertiary education statistically insignificant. The findings show that only urban access to clean cooking fuels and technologies contributes to gender equality in primary education through female employment. These findings also confirm that female employment is not a transmission channel through which access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, and urban electrification affect gender equality in secondary and tertiary education.

Evidence from Table 8 shows that mobile phone subscriptions have a statistically significant positive effect on the gender parity index for primary education but have a statistically significant negative relationship with the gender parity index for secondary and primary education. Also, in Table 8, Panel A shows that the inclusion of mobile phones as an independent variable does not either reduce the coefficients or render the impact of access to clean cooking fuels and

Table 8
Analysis of potential channels: energy poverty, mobile phone and gender equality in education [two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12
	Gender Parity Index for Primary School				Gender Parity Index for Secondary School				Gender Parity Index for Tertiary School			
Mobile phone subscription	0.009***	0.007***	0.003	0.006**	−0.006	−0.007	−0.016***	−0.007	−0.049***	−0.051***	−0.060***	−0.054***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.006)	(0.006)	(0.006)	(0.006)	(0.012)	(0.012)	(0.010)	(0.011)
Access to clean cooking fuels and technologies (urban)	−0.003				0.023***				0.050***			
	(0.004)				(0.009)				(0.015)			
Electrification		0.033***				0.086***				0.138***		
		(0.010)				(0.022)				(0.036)		
Rural electrification			0.015**				0.043***				0.068***	
			(0.007)				(0.014)				(0.021)	
Urban electrification				0.106***				0.286***				0.387***
				(0.016)				(0.040)				(0.077)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.391***	−0.201*	−0.129	−0.459***	−0.537**	−0.688***	−0.673**	−1.669***	−1.235**	−1.480***	−1.527***	−2.736***
	(0.123)	(0.110)	(0.115)	(0.108)	(0.237)	(0.256)	(0.264)	(0.245)	(0.486)	(0.486)	(0.431)	(0.456)
Observations	661	663	596	656	558	560	507	555	503	504	457	501
R^2	0.330	0.355	0.290	0.394	0.480	0.504	0.440	0.550	0.641	0.649	0.668	0.668
Hansen	14.049	2.477	0.184	0.221	2.598	2.326	0.077	0.072	2.466	0.295	0.298	0.004
Hansen (p-value)	0.000	0.115	0.668	0.639	0.107	0.127	0.782	0.789	0.116	0.587	0.585	0.947
F-statistics	46,159.389	3940.672	605.949	746.246	34,722.660	3191.978	529.727	756.626	31,054.281	2972.131	545.894	362.278
No. of instruments	11	11	11	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98
Panel B [Initial results for Comparison, see Table 2]												
Access to clean cooking fuels and technologies (urban)	**−0.002**				**0.024*** **				**0.052*** **			
	(0.004)				**(0.009)**				**(0.016)**			
Electrification		**0.033*** **				**0.084*** **				**0.119*** **		
		(0.010)				**(0.021)**				**(0.037)**		
Rural electrification			**0.015** **				**0.040*** **				**0.061*** **	
			(0.007)				**(0.013)**				**(0.022)**	
Urban electrification				**0.105*** **				**0.283*** **				**0.333*** **
				(0.015)				**(0.040)**				**(0.082)**

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

Table 9
Analysis of potential channels: Energy poverty, ICT goods and Gender equality in Education [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	Gender Parity Index for Primary School					Gender Parity Index for Secondary School					Gender Parity Index for Tertiary School				
ICT goods	−0.001	−0.001	−0.002	−0.002	−0.001	−0.013*	−0.013*	−0.012	−0.027***	−0.011	−0.030*	−0.028*	−0.030*	−0.010	−0.027
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.007)	(0.007)	(0.008)	(0.006)	(0.008)	(0.016)	(0.015)	(0.017)	(0.015)	(0.017)
Access to clean cooking fuels and technologies	**−0.001**					**0.030*** **					**0.063*** **				
	(0.004)					**(0.009)**					**(0.016)**				
Access to clean cooking fuels and technologies (Urban)		**−0.003**					**0.034*** **					**0.060*** **			
		(0.004)					**(0.009)**					**(0.016)**			
Electrification			**0.006**					**0.027**					**0.081** **		
			(0.012)					**(0.023)**					**(0.034)**		
Rural electrification				**0.006**					**0.033** **					**0.053** **	
				(0.007)					**(0.013)**					**(0.022)**	
Urban electrification					**0.002**					**0.109** **					**0.190***
					(0.029)					**(0.052)**					**(0.104)**
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.013	−0.054	0.057	0.047	0.047	−0.099	−0.108	−0.505*	−0.382	−0.958***	−0.528	−0.707	−1.290**	−1.465***	−2.132***
	(0.111)	(0.109)	(0.101)	(0.107)	(0.131)	(0.254)	(0.248)	(0.284)	(0.293)	(0.315)	(0.614)	(0.585)	(0.577)	(0.505)	(0.636)
Observations	607	607	609	558	602	517	517	519	476	514	470	470	471	435	468
R^2	0.257	0.258	0.254	0.228	0.253	0.467	0.471	0.453	0.414	0.461	0.627	0.626	0.618	0.651	0.621
Hansen	10.088	13.713	1.123	0.548	0.220	0.230	0.846	1.517	0.050	0.075	0.018	0.394	0.334	0.501	0.109
Hansen (p-value)	0.001	0.000	0.289	0.459	0.639	0.631	0.358	0.218	0.823	0.784	0.893	0.530	0.563	0.479	0.742
F-statistics	24,409.684	39,836.943	3373.353	545.933	290.859	19,203.208	30,112.380	3430.377	509.507	1712.201	16,517.110	26,554.014	2570.530	472.867	1367.082
No. of instruments	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98
Panel B [Initial results for Comparison, see Table 2]															
Access to clean cooking fuels and technologies	**0.000**					**0.024*** **					**0.052*** **				
	(0.004)					**(0.009)**					**(0.016)**				
Access to clean cooking fuels and technologies (Urban)		**−0.002**					**0.022** **					**0.048*** **			
		(0.004)	**0.033*** **				**(0.009)**	**0.084*** **				**(0.016)**	**0.119*** **		
Electrification			**(0.010)**					**(0.021)**					**(0.037)**		
				0.015 **					**0.040*** **					**0.061*** **	
Rural electrification				**(0.007)**					**(0.013)**					**(0.022)**	
					0.105* **					**0.283*** **					**0.333*** **
Urban electrification					**(0.015)**					**(0.040)**					**(0.082)**

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

Table 10
Analysis of potential channels: Energy poverty, telephone penetration and Gender equality in Education [Two-step GMM results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Telephone penetration	0.012***	0.002	0.013***	0.007**	0.004	0.003	0.001	−0.014**	0.004	−0.002	−0.008	−0.006	0.017	−0.008	0.017	0.015	0.001	0.009
	(0.003)	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)	(0.007)	(0.006)	(0.007)	(0.007)	(0.007)	(0.006)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
Access to clean cooking fuels and technologies	−0.008**						0.024**						0.040**					
	(0.004)						(0.009)						(0.018)					
Access to clean cooking fuels and technologies (Rural)		0.011***						0.054***						0.108***				
		(0.004)						(0.010)						(0.020)				
Access to clean cooking fuels and technologies (Urban)			−0.011***						0.021**						0.034*			
			(0.004)						(0.010)						(0.018)			
Electrification				0.023**						0.082***						0.104**		
				(0.010)						(0.025)						(0.043)		
Rural electrification					0.010						0.042***						0.058**	
					(0.006)						(0.015)						(0.024)	
Urban electrification						0.098***						0.298***						0.341***
						(0.018)						(0.048)						(0.100)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−0.338***	0.052	−0.381***	−0.189*	−0.178	−0.461***	−0.536**	−0.130	−0.533**	−0.742***	−0.783***	−1.747***	−0.830	−0.166	−0.967*	−1.212**	−1.486***	−2.453***
	(0.122)	(0.107)	(0.117)	(0.106)	(0.111)	(0.113)	(0.240)	(0.222)	(0.235)	(0.262)	(0.263)	(0.276)	(0.536)	(0.535)	(0.516)	(0.515)	(0.466)	(0.570)
Observations	653	645	653	654	594	647	557	552	557	558	508	553	498	495	498	499	456	496
R^2	0.333	0.309	0.338	0.340	0.292	0.374	0.476	0.498	0.474	0.492	0.432	0.538	0.620	0.644	0.619	0.624	0.637	0.639
Hansen	1.483	0.858	10.386	3.134	0.107	0.331	0.017	1.590	2.183	2.396	0.136	0.067	0.121	0.548	1.030	0.436	0.357	0.058
Hansen (p-value)	0.223	0.354	0.001	0.077	0.744	0.565	0.896	0.207	0.139	0.122	0.712	0.795	0.728	0.459	0.310	0.509	0.550	0.809
F-statistics	29,220.561	17,778.194	42,154.891	3327.475	577.376	763.508	23,793.102	15,641.995	32,471.058	2898.059	511.190	539.015	19,565.988	13,542.466	28,803.259	2586.932	557.451	413.607
No. of instruments	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
No. of countries	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98	98
Panel B [Initial results for Comparison, see Table 2]																		
Access to clean cooking fuels and technologies	0.000						0.024***						0.052***					
	(0.004)						(0.009)						(0.016)					
Access to clean cooking fuels and technologies (rural)		0.013***						0.048***						0.106***				
		(0.004)						(0.009)						(0.018)				

(*continued on next page*)

Table 10 (*continued*)

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to clean cooking fuels and technologies (urban)			−0.002						0.022**						0.048***			
			(0.004)						(0.009)						(0.016)			
Electrification				0.033***						0.084***						0.119***		
				(0.010)						(0.021)						(0.037)		
Rural electrification					0.015**						0.040***						0.061***	
					(0.007)						(0.013)						(0.022)	
Urban electrification						0.105***						0.283***						0.333***
						(0.015)						(0.040)						(0.082)

Heteroscedasticity robust standard errors in parentheses. F-statistics is the Kleibergen-Paap rk Wald F statistic for weak instrument identification. The null hypothesis that the instrument is weakly correlated with energy access variables (endogenous variables) is strongly rejected by the Kleibergen-Paap rk Wald F statistic since their values exceed Stock-Yogo weak ID test critical values at 10 %. The probability value for the Hansen J-statistics suggests that instruments are not over-identified.

* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

technologies and urban electrification on the gender parity index for primary, secondary, and tertiary education statistically insignificant. This finding shows that mobile phones, as one of the indicators of ICT, do not provide the potential mechanisms through which access to energy affects gender equality in education at all levels.

In Table 9, ICT goods have a significant negative relationship with the gender parity index for primary, secondary, and tertiary education, indicating that ICT goods exported are not gender equality enhancing. Evidence from Table 8, Panel A shows that besides access to clean cooking fuels and technologies, the inclusion of ICT goods as an additional variable renders the impact of urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification on the gender parity index for primary education statistically insignificant. Also, besides access to clean cooking fuels and technologies and urban access to clean cooking fuels and technologies, the inclusion of personal computers as an additional control variable reduces the size of the coefficients of the impact of electrification, rural electrification, and urban electrification on the gender parity index for secondary and tertiary education. These findings show that ICT goods serve as potential transmission channels through which access to energy and, more specifically, electricity affect gender equality in education.

In Table 10, telephone penetration has a statistically significant positive effect on the gender parity index for primary education but largely has an insignificant effect on the gender parity index for secondary and primary education. In Table 10, Panel A indicates that besides access to clean cooking fuels and technologies and urban access to clean cooking fuels and technologies, the inclusion of the telephone as an additional control variable reduces the size of the coefficients of urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification on the gender parity index for primary education. Further, the evidence suggests that the inclusion of telephone as an additional independent variable only reduces the size of the coefficients of urban access to clean cooking fuels and technologies and electrification on the gender parity index for secondary education. Also, besides rural access to clean cooking fuels and technologies, the inclusion of the telephone as an additional control variable reduces the size of the coefficients of the impact of access to clean cooking fuels and technologies, urban access to clean cooking fuels and technologies, electrification, rural electrification, and urban electrification on the gender parity index for tertiary education. These findings confirm that telephone penetration serves as a potential transmission channel through which access to energy improves gender equality in education at all levels.

4.3. Robustness check using alternative IV technique: Kinky and Lewbel 2SLS estimators

In addition to the two-step GMM estimator, we applied two instrumental variable techniques that could address endogeneity. The first technique applied to address endogeneity and further test the robustness of the two-step GMM estimator results is the Kinky and Lewbel 2SLS estimators to ascertain the robustness of the two-step GMM estimator results. The Kinky least squares estimator is an instrument-free estimator that addresses endogeneity by imposing assumptions on the degree of regressor endogeneity (Kripfganz and Kiviet, 2021). The Kinky least squares regression technique provides narrower confidence intervals for their estimates relative to the traditional two-stage least squares estimators (Kripfganz and Kiviet, 2021). The Kinky least squares estimator results are presented in Table 11, and the results are consistent with the two-step GMM estimator results. From Table 11, it is obvious that access to clean cooking fuels and technologies positively affects the gender parity index for primary, secondary, and tertiary education. Also, rural populations' access to clean cooking fuels and technologies and urban populations' access to clean cooking fuels and technologies have a positive and statistically significant impact on the gender parity index for primary, secondary, and tertiary education. Also, electrification has a

consistently significant positive effect on the gender parity index for primary, secondary, and tertiary education. Also, rural and urban electrification has a positive and statistically significant effect on the gender parity index for primary, secondary, and tertiary education.

In addition, the second IV technique used to address endogeneity and further test the robustness of the two-step GMM estimator results in the Lewbel two-stage least squares (Lewbel 2SLS) estimator. The Lewbel 2SLS estimator is applicable to situations when it is challenging to find an appropriate external instrument or when the external instrument is weak enough to identify the structural equation (Lewbel, 2012). In view of this, the Lewbel 2SLS estimator internally generates its instruments, which are heteroskedasticity-based, by multiplying the auxiliary equation residuals and each of the included exogenous variables in mean-centred form (Lewbel, 2012). The Lewbel 2SLS estimator does not need to satisfy the exclusion restrictions. This estimator has been applied to address endogeneity in a number of studies (see, for instance, Chaudhry and Shafiullah, 2021; Churchill and Smyth, 2020). Table 12 also presents the Lewbel 2SLS results. The Lewbel 2SLS results are not qualitatively different from the two-step GMM estimator results in estimator results. As presented in Table 12, access to clean cooking fuels and technologies has a positive and statistically significant effect on the gender parity index for primary, secondary, and tertiary education. Similarly, rural populations' access to clean cooking fuels and technologies and urban populations' access to clean cooking fuels and technologies have a positive and statistically significant impact on the gender parity index for primary, secondary, and tertiary education. Consistently, electrification has a positive and statistically significant impact on the gender parity index for primary, secondary, and tertiary education. Also, rural and urban electrification has a positive and statistically significant effect on the gender parity index for primary, secondary, and tertiary education.

4.4. Robustness check with Driscoll-Kraay and FGLS estimators

This section futher tests the robustness of two-step GMM results using Driscoll-Kraay and FGLS estimators. We present Driscoll-Kraay's results in Table 13 while presenting FGLS results in Table 14. As presented in Table 13, Driscoll-Kraay's results show evidence that access to clean cooking fuels and technologies has an insignificant effect on the gender parity index for primary and secondary education [see Columns 1 and 7]. However, in Column 13, access to clean cooking fuels and technologies has a statistically significant positive effect on the gender parity index for tertiary education at a 1 % level. Disaggregating access to clean cooking fuels and technologies into rural and urban access, we observe that rural access to clean cooking fuels and technologies has a positive association with the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Column 2, 8 and 14]. While urban access to clean cooking fuels and technologies has a statistically insignificant effect on the gender parity index for primary and secondary education, it has a statistically significant positive effect on the gender parity index for tertiary education at a 1 % level. The estimated coefficients show that access to electricity has a more substantial positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 4, 10, and 16]. Similarly, rural electrification has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 5, 11, and 17]. Urban electrification has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 6, 12, and 18].

In Table 14, the FGLS estimator shows evidence that access to clean cooking fuels and technologies positively affects the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 1, 7, and 13]. Rural access to clean cooking fuels and technologies positively correlates with the gender parity index for primary, secondary, and tertiary education, but the impact is statistically significant for

Table 11
Energy poverty and Gender equality in Education [Kinky regression results].*,**

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to clean cooking fuels and technologies	0.175***						0.347***						0.721***					
	(0.048)**						(0.077)						(0.201)*					
Access to clean cooking fuels and technologies (rural)		0.033***						0.084***						0.198***				
		(0.004)						(0.009)						(0.017)				
Access to clean cooking fuels and technologies (urban)			0.092***						0.199***						0.411***			
			(0.013)						(0.024)						(0.058)			
Electrification				0.281***						0.666***						1.138***		
				(0.044)						(0.114)						(0.203)		
Rural electrification					0.141***						0.327***						0.495***	
					(0.023)						(0.053)						(0.082)	
Urban electrification						0.337***						0.850***						1.387***
						(0.029)						(0.077)						(0.158)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	716	705	716	718	663	711	601	596	601	603	560	598	539	534	539	540	502	537

Standard errors in parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

Table 12
Energy poverty and gender equality in education [Lewbel 2SLS results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to clean cooking fuels and technologies	0.006*						0.042***						0.071***					
	(0.003)						(0.007)						(0.014)					
Access to clean cooking fuels and technologies (rural)		0.014***						0.055***						0.103***				
		(0.003)						(0.007)						(0.014)				
Access to clean cooking fuels and technologies (urban)			0.002						0.039***						0.055***			
			(0.004)						(0.008)						(0.013)			
Electrification				0.038***						0.068***						0.111***		
				(0.008)						(0.016)						(0.025)		
Rural electrification					0.016***						0.028**						0.031*	
					(0.006)						(0.011)						(0.018)	
Urban electrification						0.124***						0.274***						0.377***
						(0.014)						(0.031)						(0.059)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.132	0.133	−0.146	−0.273**	−0.157	−0.503***	−0.005	0.026	−0.093	−0.377*	−0.517**	−1.762***	0.055	−0.173	−0.186	−0.978**	−1.309***	−2.119***
	(0.112)	(0.102)	(0.109)	(0.106)	(0.108)	(0.108)	(0.220)	(0.208)	(0.216)	(0.214)	(0.232)	(0.237)	(0.454)	(0.435)	(0.452)	(0.428)	(0.418)	(0.425)
Observations	663	652	663	665	598	658	560	555	560	562	509	557	504	499	504	505	458	502
R2	0.321	0.305	0.317	0.350	0.291	0.385	0.469	0.494	0.466	0.482	0.410	0.547	0.623	0.648	0.621	0.625	0.632	0.637
widstat	32,084.074	8137.804	43,256.631	1535.877	262.792	694.026	23,552.846	7331.061	29,155.474	1491.167	310.447	693.764	14,658.938	6109.508	32,358.881	1164.072	243.779	300.777

Standard errors in parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

Table 13
Energy poverty and gender equality in education [Driscoll-Kraay results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to clean cooking fuels and technologies	−0.001						0.022						0.051***					
	(0.005)						(0.014)						(0.012)					
Access to clean cooking fuels and technologies (rural)		0.010***						0.037***						0.110***				
		(0.002)						(0.009)						(0.017)				
Access to clean cooking fuels and technologies (urban)			−0.002						0.023						0.044***			
			(0.006)						(0.019)						(0.009)			
Electrification				0.036***						0.095***						0.135***		
				(0.008)						(0.012)						(0.030)		
Rural electrification					0.013***						0.043***						0.046**	
					(0.003)						(0.011)						(0.017)	
Urban electrification						0.096***						0.288***						0.336***
						(0.008)						(0.035)						(0.025)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−0.332***	−0.004	−0.358***	−0.224	−0.232	−0.466***	−0.742*	−0.417	−0.759**	−0.802	−0.863	−1.738***	−0.971**	−0.213	−1.150**	−1.336***	−1.711***	−2.454***
	(0.089)	(0.193)	(0.085)	(0.143)	(0.195)	(0.122)	(0.359)	(0.761)	(0.296)	(0.571)	(0.697)	(0.491)	(0.392)	(0.521)	(0.450)	(0.335)	(0.501)	(0.402)
Observations	716	705	716	718	663	711	601	596	601	603	560	598	539	534	539	540	502	537
R^2	0.372	0.359	0.373	0.396	0.360	0.432	0.496	0.503	0.498	0.519	0.482	0.569	0.618	0.642	0.617	0.624	0.628	0.639

Standard errors in parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

the gender parity index for secondary and tertiary education at a 1 % level [see Columns 8 and 14]. Urban access to clean cooking fuels and technologies has a statistically significant positive effect on the gender parity index for primary, secondary, and secondary education at a 1 % level [see Columns 3, 9, and 15]. The estimated coefficients show that access to electricity has a more substantial positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 4, 10, and 16]. Rural electrification has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 5, 11, and 17]. Urban electrification has a statistically significant positive effect on the gender parity index for primary, secondary, and tertiary education at a 1 % level [see Columns 6, 12, and 18].

5. Discussion and policy implications

In this study, we examine whether access to energy contributes to gender equality in education for a panel of 98 countries between 2000 and 2021. We deployed the two-step generalized method of moments technique to account for endogeneity while Kinky least squares, Lewbel 2SLS, Driscoll-Kraay and FGLS estimators were used as alternative estimation techniques for robustness checks. Although our study examines the direct impact of gender equality in education, we also contribute to the literature by exploring the potential pathways through which access to energy contributes to gender equality in education. A mediation analysis involving a two-stage approach was followed to identify the potential channels through which access to energy influences gender equality in education.

Evidence from the two-step generalized method of moments technique, consistent with the Driscoll-Kraay and FGLS estimator, shows that energy poverty reduction contributes to gender equality in education. Specifically, this study contributes to knowledge and practice by showing that access to clean cooking fuels and technologies is associated with improvement in gender equality in secondary and tertiary education. Also, access to electricity enhances gender equality in primary, secondary, and tertiary education. Our results align with Nguyen and Su (2021) findings, which suggested that access to energy improves gender equality in education in their sample. However, in three (3) main ways, our findings differ from those of Nguyen and Su (2021). First, our estimates show that the impact of access to electricity and clean cooking technologies on gender equality in education increases at higher levels of education. In other words, access to electricity and clean cooking technologies, in descending order, has a higher impact on gender equality in tertiary education, followed by secondary and primary education. Second, while Nguyen and Su (2021) found that access to clean cooking fuels and technologies worsens gender inequality in tertiary education, we found that clean cooking fuels and technologies improve gender equality in tertiary education. Third, our results depict that, on average, the impact of access to electricity on gender equality in education outweighs the impact of access to clean cooking fuels and technologies, reflecting the unequal access to electricity and clean cooking fuels and technologies for cooking among people.

Further, our findings contribute to knowledge and policy by revealing that rural and urban access to clean cooking fuels and technologies and rural and urban electrification enhance gender equality in primary, secondary, and tertiary education. However, the impact of urban electrification and urban access to clean cooking technologies on gender equality in education outweighs the effect of rural electrification and rural access to clean cooking technologies on gender equality. This highlights the urban-rural divide in access to energy infrastructure and services. For instance, the International Energy Agency (2021) report shows that in 2010, about 97 % of the global urban population had access to electricity, while 87 % of the global rural population had access to electricity. Based on our sample, the evidence from the World Bank WDI shows that in SSA, 28.5 % of the rural population has access to electricity, while 78.2 % of urban population had access to electricity in

Table 14
Energy poverty and gender equality in education [FGLS results].

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Gender Parity Index for Primary School						Gender Parity Index for Secondary School						Gender Parity Index for Tertiary School					
Access to clean cooking fuels and technologies	0.015***						0.056***						0.096***					
	(0.004)						(0.008)*						(0.015)					
Access to clean cooking fuels and technologies (rural)		0.004						0.029***						0.114***				
		(0.003)						(0.006)						(0.012)				
Access to clean cooking fuels and technologies (urban)			0.020***						0.053***						0.088***			
			(0.004)						(0.008)						(0.017)			
Electrification				0.036***						0.112***						0.170***		
				(0.006)						(0.011)						(0.026)		
Rural electrification					0.005**						0.023***						0.044***	
					(0.002)						(0.005)						(0.012)	
Urban electrification						0.059***						0.216***						0.222***
						(0.010)						(0.026)						(0.053)
Control variables	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Constant	−0.189***	−0.210***	−0.222***	−0.287***	−0.235***	−0.455***	−0.377***	−0.348***	−0.572***	−0.850***	−0.616***	−1.333***	−1.946***	−1.276***	−2.300***	−2.622***	−2.507***	−3.267***
	(0.045)	(0.051)	(0.040)	(0.039)	(0.039)	(0.051)	(0.085)	(0.102)	(0.083)	(0.096)	(0.089)	(0.114)	(0.203)	(0.227)	(0.185)	(0.185)	(0.204)	(0.280)
Observations	712	701	712	712	655	705	599	594	599	599	555	595	536	530	536	536	496	533

Standard errors in parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

2020. In Latin America and the Caribbean, >93.5 % of the rural population has access to electricity, while >99 % of the urban population had electricity in 2020. In South Asia, >93.6 % of rural population has access to electricity, while 99.7 % of urban population had access to electricity in 2020. Also, in SSA, 5.9 % of the rural population has access to clean cooking fuels and technologies, while 35.1 % of urban population had access to clean cooking fuels and technologies in 2020. In Latin America and the Caribbean, >61.5 % of the rural population has access to clean cooking fuels and technologies, while >95.1 % of the urban population had access to clean cooking fuels and technologies in 2020. In South Asia, >44.6 % of rural population had access to clean cooking fuels and technologies, while 86.9 % of urban population had access to clean cooking fuels and technologies in 2020. Based on this information, we argue that the speed/rate at which rural and urban access to energy improves gender equality in education would differ.

Our study contributes significantly to the literature and is the first to perform pathway (mediation) analysis to explain the positive linkage between access to energy and gender equality in education. Evidence from the potential pathway (mechanism) analysis indicates multiple channels through which access to energy contributes to the enhancement of gender equality in education. First, we identified that female literacy mediates the relationship between access to energy and gender equality in education. Evidence suggests that access to electricity is associated with prolonging students' studying time in both schools and at home and further enables students to perform school activities that require electricity (Barron and Torero (2014); (Sovacool, 2012) and Khandker et al. (2014). In addition, access to electricity improves girls' school attendance and academic performance (Brew-Hammond and Crole-Rees, 2004). Also, Barkat et al. (2002) provided evidence that the literacy rate among women with access to electricity is 20 % higher than those without electricity. Barnes and Sen (2004) also showed that 90 % of women who can read have access to electricity in India. Similarly, access to clean fuels and technologies for cooking saves girls and women productive time that can be used for studying and also improve their school attendance. Our evidence, therefore, suggest that improving access to energy would improve girls and women writing and reading ability and further enahnce their academic performance as well as their retention and progression in schools, leading to improvement in gender equality in education.

Second, we also identified female health mediating the effect of access to energy on gender equality in education. In developing countries, women are seen as household energy managers (Shrestha et al., 2021). Therefore, their exposure to inefficient cooking fuels and technologies adversely impacts their health. Using firewood, charcoal, and other dirty biomass produces harmful pollutants such as carbon monoxide, nitrogen oxides, benzene, butadiene, and many other poisonous chemicals (Sovacool, 2012). These harmful pollutants are noted to be associated with stroke, ischaemic heart disease, chronic obstructive pulmonary disease, eye problems, lung cancer, cardiovascular disease, asthma, and respiratory infections, among others (Sovacool, 2012; WHO, 2021). In addition, women and girls face physical injury risks in firewood collections (Rewald, 2017; Sovacool, 2012). These adverse health effects associated with energy poverty can affect women's and girls' education through lateness to school, increased absenteeism, complete dropout, and inability to concentrate on school activities due to illness (Ozughalu and Ogwumike, 2018; Sovacool, 2012). Our findings, therefore, highlight that improving access to electricity and clean fuels and technologies for cooking improves women's and girls' health (Jayasinghe et al., 2021; Nguyen and Su, 2021; Pachauri and Rao, 2013; Pueyo and Maestre, 2019) and that could enhance their attendance, retention, academic performance and progression in schools, leading to improvement in gender equality in education.

Third, we further identified that female employment is one of the pathways through which energy access (urban access to clean cooking fuels and technologies) improves gender equality in education. In rural developing countries, girls and women spend a significant portion of their time on providing unpaid daily jobs, including collecting firewood for household activities (ENERGIA, 2018; Njoh et al., 2018). Girls and women spending much of their productive time collecting firewood can reduce their income and further encourage absenteeism and school dropout, worsening gender inequality in education. Also, the collection of fuel takes a considerable amount of households' time and hence could reduce the time available for other activities such as education (Duflo et al., 2008; Ozughalu and Ogwumike, 2018; Ssennono et al., 2021). For example, Karlsson and McDade (2001) indicate that when mothers are overburdened with performing household tasks (including searching for energy), they are more likely to make their daughters stay out of school to assist them. Lambrou and Piana (2006) and Clancy (2002) also find that girls are relatively at greater risk of being kept out of school to help collect biomass when it becomes scarce or difficult to collect. Ngarava et al. (2022) also opine that the girl child is overburdened with the household's energy provision, increasing her chances of withdrawing from school and becoming illiterate. Also, women having access to energy could improve their accumulation of assets, educational opportunities, economic opportunities, and productivity (Dinkelman, 2011; ENERGIA, 2018). Evidence shows that access to electricity increases female employment (Dinkelman, 2011). Our empirical findings confirm that through access to energy, female employment could contribute to gender equality in education in developing countries.

Finally, ICT (ICT goods exported and telephone penetration) was identified as a potential mechanism through which access to energy contributes to gender equality at all levels of education. ICT plays a significant role in education, enabling students to access information that enhances their learning (Pachauri and Rao, 2013; Rewald, 2017). Thus, with access to energy, students can acquire knowledge through television, computers, and other media (Pachauri and Rao, 2013; Rewald, 2017). In addition, evidence shows that households/women with ICT access are more likely to send girls to school (Jensen and Oster, 2009). Also, Liang et al. (2022) indicated that ICT could facilitate school girls' education pathways and opportunities for inclusive and equitable education. However, ICT cannot operate or function effectively without electricity (Armey and Hosman, 2016; Unwin and Unwin, 2009). Our findings generally indicate that for ICT to contribute to gender equality in education, policymakers must reduce energy poverty in developing countries.

6. Conclusion and direction for future research

In this study, we investigate whether access to electricity and clean cooking fuels and technologies contribute to gender equality in education for a panel of 98 countries between 2000 and 2021 using various econometric techniques. The findings from the two-step generalized method of moments technique, consistent with the Kinky least squares, Lewbel 2SLS, Driscoll-Kraay and FGLS estimators results, can be grouped into three (3) parts. First, the findings generally show that improving access to electricity and clean cooking fuels and technologies leads to gender equality at all levels of education. Second, rural and urban access to clean cooking fuels and technologies and rural and urban electrification enhance gender equality at all levels of education. However, the impact of urban electrification and urban access to clean cooking technologies on gender equality in education outweighs the effect of urban electrification and urban access to clean cooking technologies on gender equality. Finally, the mediation analysis indicates that female literacy, female health, female employment, and ICT (ICT goods exported and telephone penetration) are the potential transmission channels through which access to energy contributes to gender equality at all levels of education.

From a policy perspective, our findings highlight that access to energy can be an important policy option for addressing gender inequality in education, which remains a challenge in developing regions. Therefore, we recommend policymakers to implement policies that would facilitate the eradication of energy poverty in developing countries. In

addition, policies designed to address energy poverty in developing countries should address the urban-rural divide in access to energy. Further, policies that promote female literacy, health, employment, and ICT penetration in developing countries are essential since they serve as channels through which access to energy could enhance gender equality in education. Despite the numerous strengths of this paper, there is still an avenue for further research. First, our study addresses the contribution of energy access to gender equality in education. However, our study did not address the role of energy justice in gender inequality in education. We, therefore, recommend future studies to explore the direct and potential pathways through which energy justice affects gender equality in education. Second, while we focus on only gender equality in education, future studies could also test the effect of energy access and justice on other dimensions of gender inequality in health, employment, and social rights.

Declaration of conflict of interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Alex O. Acheampong: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eric Evans Osei Opoku:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Afua Amankwaa:** Writing – review & editing, Writing – original draft. **Janet Dzator:** Writing – review & editing, Writing – original draft.

Data availability

Data will be made available on request.

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Appendix A

Appendix Table 1
List of countries included in the study.

South Asia countries
Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka
Sub-Saharan Africa countries
Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Dem. Rep., Congo, Rep., Cote d'Ivoire, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, The, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, Sudan, Tanzania, Togo, Uganda, Zambia, Zimbabwe
Latin America & Caribbean countries
Antigua and Barbuda, Argentina, Aruba, Bahamas, The, Barbados, Belize, Bolivia, Brazil, British Virgin Islands, Cayman Islands, Chile, Colombia, Costa Rica, Cuba, Curacao, Dominica, Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Puerto Rico, Sint Maarten (Dutch part), St. Kitts and Nevis, St. Lucia, St. Martin (French part), St. Vincent and the Grenadines, Suriname, Trinidad and Tobago, Turks and Caicos Islands, Uruguay, Venezuela, RB, Virgin Islands (U.S.)

Appendix Table 2
Summary statistics.

Variable	Indicators	Mean	Std. Dev.	Min	Max
Gender parity in primary education	lnpse_pp	−0.060	0.104	−0.837	0.149
Gender parity in secondary education	lnsse_pp	−0.070	0.240	−1.563	0.343
Gender parity in tertiary education	lntss_pp	−0.190	0.539	−2.742	0.594
Access to clean cooking fuels and technologies	lnclean	2.858	1.840	−2.303	4.605
Access to clean cooking fuels and technologies (Rural)	lnclean_ru	1.997	2.279	−2.303	4.605
Access to clean cooking fuels and technologies (Urban)	lnclean_urb	3.345	1.629	−2.303	4.605
Electrification	lnelec	3.917	0.867	−0.441	4.605
Rural electrification	lnelec_ru	3.476	1.343	−0.648	4.605
Urban electrification	lnelec_urb	4.315	0.457	1.232	4.605
Access to credit	lndcp	3.050	0.919	−0.910	4.959
GDP per capita	lnrgdpc	7.925	1.312	5.555	11.423
Foreign aid	lnoda	19.283	1.657	11.156	23.198
Foreign direct investment	lnfdi	1.005	1.347	−6.684	7.444
Trade openness	lntrade	4.161	0.577	−0.243	6.761
Internet penetration	lninter	2.042	1.787	−5.390	4.576
Education expenditure	lnedu_expt	2.709	0.389	−0.182	3.625
Female literacy	lnlit_yfe	4.375	0.350	2.656	4.605
Female health	lnlifexp_fe	4.196	0.157	3.689	4.439
Female employment	lnempl_fe	0.029	1.129	−3.507	2.257
ICT goods	lnicts_exp	1.498	1.171	−4.820	4.118

(*continued on next page*)

Appendix Table 2 (*continued*)

Variable	Indicators	Mean	Std. Dev.	Min	Max
Mobile phone subscription	lnmsc	14.307	2.421	7.090	20.906
Telephone penetration	lnfts	11.946	2.087	5.011	17.728

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